

Curated Results, Tables, and Figure Compendium

Companion to Manuscript v9 — A recurrent transcriptomic context for distal tubule transporter suppression in mouse spaceflight kidney

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This document collects the curated result narrative, manuscript tables, supplementary tables, main figures, and figure gallery supporting Manuscript v9. It follows the v9 claim hierarchy: the original RRRM-2 age-rewiring claim fails at primary resolution; terminal-flight kidney samples define a DCT/NCC-low transcriptomic state embedded in ECM/stromal, S1P/S1PR3, integrin/adhesion, MMP/ADAM proteolysis, TLR/innate/macrophage, and preservation/stress-response programs; live return attenuates or redirects this state; OSD-253 supports TLR4/innate association rather than TLR4 requirement; and OSD-462 anchors only the transporter side of the state to the established NCC/SPAK phospho-suppression phenotype. The compendium does *not* claim NCC dephosphorylation as a new discovery; that phenotype was established by the Cosmic Kidney Disease study. All numbers are drawn from the run artifacts `run_20260519_000547_2500g`, `run_20260519_cosine_perm_null`, `run_20260519_5000g_sensitivity_3seed`, and `run_20260521_osd462_anchor`.

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S1 Claim Gates and Hypothesis Outcomes

The OSD-462 multi-omic anchor was specified in a pre-registered plan (docs/osd462_multiomics_analysis_plan.md) with four hypotheses, each carrying an explicit falsification rule declared before any protein or phosphoprotein effect was computed. Table S1 records the outcomes. Three hypotheses were falsified and one was strongly supported as a positive-control anchor; under the pre-registration, every outcome — including the nulls — is a reportable result.

Table S1: Pre-registered hypotheses and outcomes for the OSD-462 multi-omic anchor.

ID	Hypothesis and pre-registered prediction	Outcome
H1	Matrix/ECM transcripts and proteins are concordant in flight (matrix proteins up; concordance exceeds matched null).	Falsified. Matrix proteins moved opposite trans-scripts.
H2	DCT/NCC-WNK transcripts and proteins are concordant (transporter proteins down; concordance exceeds null).	Falsified. No protein-abundance effect; total NCC protein flat.
H3	The WNK-SPAK/OSR1-NCC phospho-axis is suppressed in flight.	Supported as anchor. NCC activating cluster and SPAK strongly down; this reproduces the established Cosmic Kidney Disease phenotype.
H4	RRRM-2 LIONESS/node2vec network candidates are enriched for OSD-462 protein- or phosphoprotein-changing genes.	Falsified. No enrichment ($p \geq 0.23$).

Decision-table outcome. Protein-abundance concordance was null for both matrix and DCT gene sets; the DCT/NCC side of the RNA state localized to the phospho/activity layer. Per the pre-registered claim ceiling, the result is a molecular claim (transcript-state plus signaling anchor), with fibrosis/tissue deposition, urine chemistry, and causal upstream drivers remaining out of scope.

S2 Cohort design

Table S2: **Cohorts analyzed.** RRRM-2 is the factorial bridge; OSD-513 and OSD-253 are independent transcriptomic cohorts; OSD-462 (RR-10) is the multi-omic anchor and the source of the Cosmic Kidney Disease mouse kidney phosphoproteomics.

Cohort	Accession	Tissue / sex	Modalities	Role
RRRM-2	OSD-771	Kidney / female	RNA-seq	Factorial bridge: Age $\times$ Arm (ISS-T, LAR) $\times$ Flight, $n = 5/\text{cell}$
OSD-513	OSD-513	Kidney / –	RNA-seq	Independent flight-vector recurrence cohort
OSD-253	OSD-253	Kidney / –	RNA-seq	Strain/mechanism context (C57BL/6J vs C3H/HeJ)
OSD-462	OSD-462 (RR-10)	Kidney / female	RNA-seq, proteomics, phosphoproteomics	Multi-omic anchor; Cosmic Kidney Disease mouse cohort
Tabula Muris Senis	—	Kidney	scRNA-seq	External aging reference

S3 Within-RRRM-2 age-vector stability (original claim, not supported)

Table S3: **Bootstrap angular stability of RRRM-2 control-aging vectors.** Module and pathway are the primary biological resolutions; three of four arm-resolution settings failed the stability gate, so the original within-cohort age-rewiring contrast is not promoted as a claim.

Arm	Resolution	Full norm	Median cosine	Lower CI	Required lower	Passed
ISS-T	pathway	0.167	0.787	0.248	0.400	no
ISS-T	module	0.325	0.750	0.056	0.400	no
ISS-T	gene	10.152	0.746	0.545	0.200	yes
ISS-T	LIONESS-node	10.152	0.745	0.549	0.200	yes
LAR	pathway	0.221	0.890	0.345	0.400	no
LAR	module	0.353	0.774	0.493	0.400	yes
LAR	gene	9.687	0.742	0.563	0.200	yes
LAR	LIONESS-node	9.687	0.737	0.556	0.200	yes

S4 Cross-OSDR flight-vector recurrence

Table S4: **Cross-OSDR pathway-vector alignment.** OSD-513 is interpreted as directional recurrence; OSD-253 as context. p_{perm} is the directional external-label permutation sensitivity with the RRRM-2 reference vector held fixed.

Study	Scenario	Arm	Point	Median	Lower CI	Upper CI	p_{perm}
OSD-513	powered HGC	ISS-T	0.641	0.625	0.351	0.814	0.127
OSD-513	powered HGC	LAR	-0.511	-0.498	-0.726	-0.231	0.178
OSD-253	original GC	ISS-T	0.674	0.602	-0.411	0.836	0.141
OSD-253	original GC	LAR	-0.679	-0.608	-0.816	0.415	0.125
OSD-253	rerun GC	ISS-T	0.474	0.470	-0.040	0.688	0.269
OSD-253	rerun GC	LAR	-0.318	-0.309	-0.572	0.183	0.344

Table S5: **Pathway construction of the recurrent ISS-T remodeling axis.** Effects are pathway-scale FLT-minus-GC contrasts; the axis is oriented so ECM/fibrosis is positive and DCT/NCC-WNK suppression is negative.

Pathway	RRRM-2 ISS-T	OSD-513	Axis weight	Primary axis
ECM remodeling	0.949	0.801	0.622	yes
Fibrosis	0.764	0.652	0.504	yes
DCT/NCC-WNK	-0.398	-0.889	-0.452	yes
Apoptosis	0.674	0.386	0.378	yes
Oxidative stress	0.198	0.104	0.108	yes
Inflammation	0.561	-0.092	0.000	no
Ion transport	0.329	-0.762	0.000	sensitivity
Lipid metabolism	-0.320	0.156	0.000	no
Calcium handling	0.003	-0.550	0.000	no

S5 DCT/NCC-low State and Live-return Arm

Table S6: **State-space component effects (FLT-minus-GC) and arm-by-flight interactions.** Young LAR effects are shown to be underpowered and not statistically distinguishable from zero.

Component	Arm / scope	Effect	Welch p / interaction p	BH q
matrix-minus-DCT	ISS-T pooled	+2.254	0.003	0.020
matrix-minus-DCT	LAR pooled	-0.112	0.865	0.895
matrix-minus-DCT	ISS-T young	+3.204	0.005	0.025
matrix-minus-DCT	LAR young	-0.782	0.361	0.542
matrix-minus-DCT	Arm $\times$ Flight interaction	+2.366	0.016	0.027
matrix component	Arm $\times$ Flight interaction	+1.738	—	0.004
DCT transport	Arm $\times$ Flight interaction	-0.628	—	0.344
mechanism-score vector	cos(LAR, ISS-T) pooled	-0.725	CI [-0.874, -0.122]	—

S6 OSD-462 multi-omic anchor

S6.1 RNA recurrence gate (Layer 4)

The OSD-462 Space Flight minus Ground Control RNA pathway vector reproduced the RRRM-2 ISS-T DCT/NCC-low context direction with cosine 0.869 (95% bootstrap CI 0.648–0.903; bootstrap median 0.823) across 11 pathways; leave-one-pathway-out cosine ranged 0.823–0.904. The recurrence gate passed.

Table S7: **OSD-462 RNA pathway flight effects vs RRRM-2 ISS-T.** Space Flight minus Ground Control (OSD-462) and FLT-minus-GC (RRRM-2 ISS-T). These define the co-traveling RNA context around the DCT/NCC-low state.

Pathway	OSD-462 RNA effect	RRRM-2 ISS-T effect	Sign agreement
ecm_organization	0.222	0.280	1.000
fibrosis_tgfb_empt	0.149	0.025	1.000
tlr4_innate	0.075	0.095	1.000
integrin_cell_adhesion	0.119	0.087	1.000
slp_slpr3	0.123	0.106	1.000
mmp_adam_proteolysis	0.197	0.123	1.000
dct_ncc_wnk_transport	-0.130	-0.075	1.000
tubular_transport_broad	-0.048	0.042	0.000
oxidative_stress_nrf2	0.021	0.005	1.000
macrophage_inflammation	0.109	0.090	1.000
preservation_stress_response	0.157	0.293	1.000

S6.2 Protein-abundance concordance (Layer 1)

Genome-wide RRRM-2-RNA versus OSD-462-protein flight-effect correlation was negligible (Spearman 0.043). No targeted gene set was concordant in its predicted direction against the abundance- and peptide-matched random-gene-set null. The matrix/ECM set moved opposite its transcripts.

Table S8: **OSD-462 protein-abundance concordance by gene set.** Signed mean protein flight effect with the matched random-gene-set null interval. No set is concordant in its predicted direction.

Gene set	Kind	n	Predicted dir.	Signed mean effect	Matched-null interval
dct_ncc_wnk_transport	family	10	down	-0.044	[-0.052, 0.047]
ecm_organization	family	15	up	-0.107	[-0.082, 0.056]
tlr4_innate	family	8	up	-0.021	[-0.040, 0.031]
slp_slpr3	family	9	up	-0.045	[-0.055, 0.054]
tubular_transport_broad	control	15	up	0.043	[-0.061, 0.060]

S6.3 WNK–SPAK/OSR1–NCC phospho-axis (Layer 2; positive-control anchor)

The phosphoproteomics checkpoint passed (all targeted NCC, SPAK, OSR1, WNK sites quantified). Total NCC protein abundance was unchanged in flight (+0.089), so phosphosite changes reflect altered phosphorylation stoichiometry. Table S9 lists every quantified phosphosite on the WNK–SPAK/OSR1–NCC chain. The NCC N-terminal regulatory (SPAK/OSR1-target) cluster and SPAK sites are strongly suppressed; non-regulatory NCC sites are unchanged — an internal negative control against a global normalization artifact. In v9 this is interpreted as a positive-control anchor to the known transporter phenotype, not as a new discovery.

Table S9: **Complete OSD-462 WNK–SPAK/OSR1–NCC phosphosite flight effects.** Space Flight minus Ground Control, log₂ scale, plex-corrected. *n* is FL/GC channels.

Gene	Site	Kind	<i>n</i>	Effect	CI low	CI high	<i>p</i> value
Calb1	60	single	5/5	0.177	-0.016	0.370	0.067
Cul3	737	single	10/10	0.121	-4.8e-04	0.243	0.051
Nedd4l	331	single	5/5	0.077	-0.223	0.378	0.570
Nedd4l	332	single	10/10	0.002	-0.069	0.073	0.955
Nedd4l	373	single	10/10	-0.230	-0.440	-0.021	0.033
Nedd4l	480	single	10/10	-0.264	-0.383	-0.144	2.4e-04
Nedd4l	484	single	5/5	-0.412	-0.637	-0.186	0.003
Nedd4l	504	single	10/10	0.040	-0.106	0.186	0.573
Oxsr1	339	single	10/10	0.005	-0.055	0.065	0.867
Sgk1	494	single	10/10	-0.114	-0.683	0.454	0.677
Slc12a3	120	single	10/10	0.027	-0.175	0.229	0.780
Slc12a3	120;122	composite	10/10	0.191	-0.046	0.429	0.107
Slc12a3	120;124	composite	10/10	0.039	-0.164	0.243	0.688
Slc12a3	122	single	10/10	0.045	-0.168	0.257	0.665
Slc12a3	122;124	composite	10/10	0.168	-0.062	0.399	0.141
Slc12a3	124	single	10/10	0.051	-0.254	0.357	0.727
Slc12a3	53	single	10/10	-0.851	-1.257	-0.445	3.8e-04
Slc12a3	53;65	composite	10/10	-0.350	-0.496	-0.204	9.7e-05
Slc12a3	53;68	composite	5/5	-0.133	-0.758	0.493	0.638
Slc12a3	56;65	composite	5/5	-0.124	-0.560	0.312	0.530
Slc12a3	58;65	composite	10/10	-0.545	-0.836	-0.254	0.001
Slc12a3	65	single	10/10	-0.790	-1.038	-0.543	3.5e-06
Slc12a3	65;68	composite	10/10	-1.563	-2.005	-1.121	9.3e-07
Slc12a3	68	single	10/10	-0.794	-1.168	-0.420	3.3e-04
Slc12a3	89	single	10/10	-0.930	-1.305	-0.556	6.6e-05
Slc12a3	96	single	10/10	0.048	-0.092	0.189	0.480
Stk39	366	single	10/10	-0.520	-0.706	-0.334	1.7e-05
Stk39	382	single	5/5	-0.829	-1.768	0.109	0.076
Stk39	382;383	composite	10/10	-0.368	-0.579	-0.157	0.002
Stk39	383	single	10/10	-0.793	-1.160	-0.426	2.8e-04
Stk39	397	single	10/10	0.073	-0.013	0.159	0.092
Wnk1	157	single	5/5	-0.129	-0.346	0.087	0.206
Wnk1	17	single	10/10	-0.159	-0.335	0.018	0.075
Wnk1	172	single	10/10	0.014	-0.128	0.156	0.834
Wnk1	185	single	5/5	0.008	-0.252	0.269	0.942
Wnk1	2258	single	10/10	-0.200	-0.482	0.083	0.154
Wnk1	2280;2282	composite	5/5	-0.121	-0.592	0.350	0.569
Wnk1	2280;2282;2285	composite	10/10	0.129	-0.017	0.275	0.080
Wnk1	2282;2285	composite	10/10	0.135	-0.120	0.390	0.280
Wnk1	2285	single	5/5	0.022	-0.467	0.510	0.921
Wnk1	2392	single	5/5	-0.628	-0.880	-0.376	4.4e-04
Wnk1	2623	single	5/5	-0.297	-0.563	-0.030	0.033
Wnk1	2625	single	10/10	-0.362	-0.609	-0.114	0.007
Wnk4	1014	single	10/10	-0.036	-0.096	0.024	0.217
Wnk4	1025	single	10/10	0.190	0.051	0.329	0.010

Table S9 continued

Gene	Site	Kind	<i>n</i>	Effect	CI low	CI high	<i>p</i> value
Wnk4	1025;1033	composite	10/10	0.287	0.093	0.481	0.006
Wnk4	1033	single	10/10	0.290	0.097	0.484	0.006
Wnk4	125	single	10/10	-0.049	-0.273	0.175	0.649
Wnk4	639	single	5/5	-0.110	-0.465	0.246	0.496
Wnk4	783	single	5/5	0.118	-0.033	0.269	0.110
Wnk4	95	single	10/10	-0.006	-0.075	0.063	0.861

S6.4 Network-candidate translation (Layer 3)

RRRM-2 LIONESS/node2vec top-composite candidates were not enriched among OSD-462 protein-changing genes at any top-*k* threshold.

Table S10: **Network-candidate translation to OSD-462 proteins.** Enrichment of RRRM-2 network-prioritized genes among protein-changing genes vs the matched null.

Top <i>k</i>	Candidates w/ protein	Mean effect	Null median	Enrichment <i>p</i>
25	14	0.103	0.081	0.227
50	22	0.088	0.079	0.329
100	41	0.089	0.080	0.298

S7 Mechanism Context and Complete State Tables

Table S11: **Mechanism convergence summary.** Mechanisms are classified as outcome anchors, candidate upstream axes, or candidate context signatures. Association q-values are from score-versus-remodeling tests; gene-priority q-values summarize whether network-ranked genes enrich for the mechanism set.

Mechanism	Role	<i>n</i>	Best assoc. <i>q</i>	Top50 <i>q</i>	Preranked <i>q</i>	OSD-253 interp.
dct_ncc_wnk_transport	outcome_anchor	15.000	3.14e-11	1.000	0.428	inconclusive
ecm_organization	outcome_anchor	5.000	4.74e-25	1.000	0.950	tlr4_not_required
fibrosis_tgfb_emt	candidate_upstream	14.000	0.365	1.000	0.430	inconclusive,tlr4_not_required
integrin_cell_adhesion	candidate_upstream	17.000	0.001	1.000	0.950	inconclusive,tlr4_not_required
macrophage_inflammation	candidate_context	13.000	7.88e-12	1.000	0.684	NA
mmp_adam_proteolysis	candidate_upstream	10.000	2.65e-05	1.000	0.950	inconclusive,tlr4_not_required
oxidative_stress_nrf2	candidate_context	6.000	0.221	1.000	NA	NA
preservation_stress_response	candidate_context	14.000	1.53e-04	1.000	0.233	NA
s1p_s1pr3	candidate_upstream	13.000	2.96e-08	1.000	NA	NA
tlr4_innate	candidate_upstream	16.000	2.29e-28	1.000	0.950	inconclusive,tlr4_not_required
tubular_transport_broad	outcome_anchor	5.000	0.016	1.000	0.076	NA

Table S12: **Mechanism-score associations with the recurrent DCT/NCC-low state.** Full curated table from `mechanism_score_associations.tsv`; estimates are Spearman correlations for score-versus-remodeling rows or FLT-minus-GC effects for flight-effect rows.

Study	Scenario	Comparison	Mechanism	Estimate	p	q
OSD-253	all_samples	score_vs_remodeling_spearman	ecm_organization	0.801	2.88e-26	4.74e-25
OSD-253	all_samples	score_vs_remodeling_spearman	fibrosis_tgfb_emt	0.104	0.277	0.365
OSD-253	all_samples	score_vs_remodeling_spearman	tlr4_innate	0.832	6.95e-30	2.29e-28
OSD-253	all_samples	score_vs_remodeling_spearman	integrin_cell_adhesion	0.316	6.78e-04	0.001
OSD-253	all_samples	score_vs_remodeling_spearman	s1p_s1pr3	0.517	5.38e-09	2.96e-08
OSD-253	all_samples	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.159	0.093	0.162
OSD-253	all_samples	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.597	3.81e-12	3.14e-11
OSD-253	all_samples	score_vs_remodeling_spearman	tubular_transport_broad	0.164	0.085	0.155
OSD-253	all_samples	score_vs_remodeling_spearman	oxidative_stress_nrf2	-0.142	0.134	0.221
OSD-253	all_samples	score_vs_remodeling_spearman	macrophage_inflammation	0.612	7.16e-13	7.88e-12
OSD-253	all_samples	score_vs_remodeling_spearman	preservation_stress_response	0.371	5.55e-05	1.53e-04
OSD-513	powered_hgc	score_vs_remodeling_spearman	ecm_organization	0.745	8.14e-06	2.69e-05
OSD-513	powered_hgc	score_vs_remodeling_spearman	fibrosis_tgfb_emt	0.091	0.652	0.694
OSD-513	powered_hgc	score_vs_remodeling_spearman	tlr4_innate	0.123	0.540	0.636
OSD-513	powered_hgc	score_vs_remodeling_spearman	integrin_cell_adhesion	0.013	0.947	0.947
OSD-513	powered_hgc	score_vs_remodeling_spearman	s1p_s1pr3	0.221	0.268	0.365
OSD-513	powered_hgc	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.284	0.150	0.226
OSD-513	powered_hgc	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.676	1.09e-04	2.78e-04
OSD-513	powered_hgc	score_vs_remodeling_spearman	tubular_transport_broad	-0.285	0.149	0.226
OSD-513	powered_hgc	score_vs_remodeling_spearman	oxidative_stress_nrf2	-0.200	0.318	0.389
OSD-513	powered_hgc	score_vs_remodeling_spearman	macrophage_inflammation	0.212	0.289	0.367
OSD-513	powered_hgc	score_vs_remodeling_spearman	preservation_stress_response	-0.444	0.020	0.039
RRRM-2	primary	score_vs_remodeling_spearman	ecm_organization	0.644	1.12e-10	7.37e-10
RRRM-2	primary	score_vs_remodeling_spearman	fibrosis_tgfb_emt	-0.066	0.562	0.640
RRRM-2	primary	score_vs_remodeling_spearman	tlr4_innate	0.488	4.48e-06	1.85e-05
RRRM-2	primary	score_vs_remodeling_spearman	integrin_cell_adhesion	0.374	6.33e-04	0.001
RRRM-2	primary	score_vs_remodeling_spearman	s1p_s1pr3	0.451	2.75e-05	8.25e-05
RRRM-2	primary	score_vs_remodeling_spearman	mmp_adam_proteolysis	0.478	7.22e-06	2.65e-05
RRRM-2	primary	score_vs_remodeling_spearman	dct_ncc_wnk_transport	-0.560	6.66e-08	3.14e-07
RRRM-2	primary	score_vs_remodeling_spearman	tubular_transport_broad	-0.296	0.008	0.016
RRRM-2	primary	score_vs_remodeling_spearman	oxidative_stress_nrf2	0.020	0.862	0.889
RRRM-2	primary	score_vs_remodeling_spearman	macrophage_inflammation	0.150	0.185	0.265
RRRM-2	primary	score_vs_remodeling_spearman	preservation_stress_response	0.062	0.584	0.643
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	ecm_organization	0.859	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	ecm_organization	-0.545	NA	NA
OSD-513	powered_hgc	flight_minus_gc	ecm_organization	0.485	0.085	0.233
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	fibrosis_tgfb_emt	0.068	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	fibrosis_tgfb_emt	-0.185	NA	NA
OSD-513	powered_hgc	flight_minus_gc	fibrosis_tgfb_emt	0.082	0.552	0.674
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	tlr4_innate	0.602	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	tlr4_innate	-0.223	NA	NA
OSD-513	powered_hgc	flight_minus_gc	tlr4_innate	0.034	0.839	0.839
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	integrin_cell_adhesion	0.445	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	integrin_cell_adhesion	-0.118	NA	NA
OSD-513	powered_hgc	flight_minus_gc	integrin_cell_adhesion	-0.120	0.417	0.655
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	s1p_s1pr3	0.727	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	s1p_s1pr3	-0.271	NA	NA
OSD-513	powered_hgc	flight_minus_gc	s1p_s1pr3	0.338	0.006	0.032
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	mmp_adam_proteolysis	0.535	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	mmp_adam_proteolysis	-0.139	NA	NA
OSD-513	powered_hgc	flight_minus_gc	mmp_adam_proteolysis	0.118	0.291	0.583
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	dct_ncc_wnk_transport	-0.398	NA	NA

RRRM-2	primary	LAR_age_balanced_flight_minus_gc	dct_ncc_wnk_transport	-0.149	NA	NA
OSD-513	powered_hgc	flight_minus_gc	dct_ncc_wnk_transport	-0.889	8.94e-04	0.010
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	tubular_transport_broad	-0.267	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	tubular_transport_broad	-0.167	NA	NA
OSD-513	powered_hgc	flight_minus_gc	tubular_transport_broad	-0.190	0.522	0.674
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	oxidative_stress_nrf2	-0.047	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	oxidative_stress_nrf2	0.271	NA	NA
OSD-513	powered_hgc	flight_minus_gc	oxidative_stress_nrf2	-0.044	0.810	0.839
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	macrophage_inflammation	0.380	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	macrophage_inflammation	-0.135	NA	NA
OSD-513	powered_hgc	flight_minus_gc	macrophage_inflammation	-0.214	0.318	0.583
RRRM-2	primary	ISS-T_age_balanced_flight_minus_gc	preservation_stress_response	0.594	NA	NA
RRRM-2	primary	LAR_age_balanced_flight_minus_gc	preservation_stress_response	-0.148	NA	NA
OSD-513	powered_hgc	flight_minus_gc	preservation_stress_response	-0.537	0.055	0.202

Table S13: **Complete state-space group effects.** Full curated component-effect table from `state_space_group_effects.tsv`. DCT transport is in native direction, so negative values indicate reduced DCT/NCC-WNK signature in flight.

Study	Scenario	Contrast	Arm	Age	Strain	Component	Effect	p	q
RRRM-2	primary	ISS-T_age.pooled_FLT_minus_GC	ISS-T	pooled	NA	matrix.component	1.250	0.002	0.016
RRRM-2	primary	ISS-T_age.pooled_FLT_minus_GC	ISS-T	pooled	NA	dct.transport.component	-1.004	0.035	0.102
RRRM-2	primary	ISS-T_age.pooled_FLT_minus_GC	ISS-T	pooled	NA	immune.context.component	1.158	8.78e-05	0.001
RRRM-2	primary	ISS-T_age.pooled_FLT_minus_GC	ISS-T	pooled	NA	preservation.stress.component	1.062	0.007	0.031
RRRM-2	primary	ISS-T_age.pooled_FLT_minus_GC	ISS-T	pooled	NA	matrix.minus.dct	2.254	0.003	0.016
RRRM-2	primary	ISS-T_YNG_FLT_minus_GC	ISS-T	YNG	NA	matrix.component	1.788	0.016	0.058
RRRM-2	primary	ISS-T_YNG_FLT_minus_GC	ISS-T	YNG	NA	dct.transport.component	-1.416	0.022	0.072
RRRM-2	primary	ISS-T_YNG_FLT_minus_GC	ISS-T	YNG	NA	immune.context.component	0.930	0.002	0.016
RRRM-2	primary	ISS-T_YNG_FLT_minus_GC	ISS-T	YNG	NA	preservation.stress.component	0.689	0.295	0.477
RRRM-2	primary	ISS-T_YNG_FLT_minus_GC	ISS-T	YNG	NA	matrix.minus.dct	3.204	0.005	0.025
RRRM-2	primary	ISS-T_OLD_FLT_minus_GC	ISS-T	OLD	NA	matrix.component	0.712	0.095	0.229
RRRM-2	primary	ISS-T_OLD_FLT_minus_GC	ISS-T	OLD	NA	dct.transport.component	-0.593	0.431	0.586
RRRM-2	primary	ISS-T_OLD_FLT_minus_GC	ISS-T	OLD	NA	immune.context.component	1.385	0.005	0.025
RRRM-2	primary	ISS-T_OLD_FLT_minus_GC	ISS-T	OLD	NA	preservation.stress.component	1.435	0.013	0.052
RRRM-2	primary	ISS-T_OLD_FLT_minus_GC	ISS-T	OLD	NA	matrix.minus.dct	1.304	0.215	0.408
RRRM-2	primary	LAR_age.pooled_FLT_minus_GC	LAR	pooled	NA	matrix.component	-0.488	0.120	0.265
RRRM-2	primary	LAR_age.pooled_FLT_minus_GC	LAR	pooled	NA	dct.transport.component	-0.377	0.461	0.604
RRRM-2	primary	LAR_age.pooled_FLT_minus_GC	LAR	pooled	NA	immune.context.component	-0.424	0.229	0.420
RRRM-2	primary	LAR_age.pooled_FLT_minus_GC	LAR	pooled	NA	preservation.stress.component	-0.265	0.540	0.632
RRRM-2	primary	LAR_age.pooled_FLT_minus_GC	LAR	pooled	NA	matrix.minus.dct	-0.112	0.865	0.904
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	matrix.component	-0.374	0.435	0.586
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	dct.transport.component	0.408	0.430	0.586
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	immune.context.component	-0.177	0.619	0.695
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	preservation.stress.component	-0.104	0.895	0.912
RRRM-2	primary	LAR_YNG_FLT_minus_GC	LAR	YNG	NA	matrix.minus.dct	-0.782	0.361	0.552
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	matrix.component	-0.603	0.198	0.389
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	dct.transport.component	-1.161	0.131	0.277
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	immune.context.component	-0.672	0.280	0.466
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	preservation.stress.component	-0.426	0.387	0.575
RRRM-2	primary	LAR_OLD_FLT_minus_GC	LAR	OLD	NA	matrix.minus.dct	0.558	0.533	0.632
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	matrix.component	0.248	0.513	0.627
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	dct.transport.component	-1.566	8.94e-04	0.008
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	immune.context.component	-0.109	0.754	0.829
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	preservation.stress.component	-0.900	0.055	0.151
OSD-513	powered_hgc	OSD-513_FLT_minus_GC	NA	pooled	NA	matrix.minus.dct	1.814	0.007	0.031
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix.component	0.749	0.035	0.102
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	dct.transport.component	-0.041	0.921	0.921
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	immune.context.component	0.500	0.248	0.426
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	preservation.stress.component	0.092	0.851	0.904
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix.minus.dct	0.789	0.096	0.229
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix.component	0.339	0.089	0.229
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	dct.transport.component	0.627	0.195	0.389
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	immune.context.component	0.077	0.871	0.904
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	preservation.stress.component	0.235	0.513	0.627
OSD-253	original_gc.blue.light	OSD-253_original_gc.blue.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix.minus.dct	-0.289	0.552	0.633
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix.component	1.641	4.72e-05	8.65e-04
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	dct.transport.component	-1.036	0.022	0.072
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	immune.context.component	0.658	0.116	0.265
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	preservation.stress.component	-0.388	0.437	0.586
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C57BL/6J_FLT_minus_GC	NA	pooled	C57BL/6J	matrix.minus.dct	2.678	8.50e-06	2.34e-04
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix.component	1.509	5.20e-08	2.86e-06
OSD-253	rerun_gc.white.light	OSD-253_rerun_gc.white.light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	dct.transport.component	-0.310	0.479	0.612

OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	immune_context_component	0.265	0.335	0.526
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	preservation_stress_component	-0.409	0.240	0.426
OSD-253	rerun_gc_white_light	OSD-253_rerun_gc_white_light_C3H/HeJ_FLT_minus_GC	NA	pooled	C3H/HeJ	matrix_minus_dct	1.819	2.84e-04	0.003

Table S14: **OSD-253 TLR4/strain discriminator.** C57BL/6J and C3H/HeJ effects by score and control scenario. The key boundary is association rather than strict TLR4 requirement.

Scenario	Score	C57 effect	C3H effect	C3H-C57	q C57	q C3H	q delta	Interp.
original_gc_blue_light	remodeling_score	0.816	0.328	-0.488	0.279	0.757	0.748	inconclusive
original_gc_blue_light	tlr4_innate	0.184	0.231	0.048	0.460	0.521	0.870	tlr4_not_required
original_gc_blue_light	fibrosis_tgfb.emt	0.190	-0.038	-0.227	0.279	0.757	0.550	inconclusive
original_gc_blue_light	ecm_organization	0.507	0.398	-0.110	0.279	0.308	0.870	tlr4_not_required
original_gc_blue_light	integrin_cell_adhesion	0.336	0.187	-0.148	0.182	0.308	0.748	inconclusive
original_gc_blue_light	mmp_adam_proteolysis	0.316	-0.051	-0.366	0.279	0.757	0.550	inconclusive
original_gc_blue_light	dct_ncc_wnk_transport	-0.030	0.214	0.244	0.791	0.463	0.550	inconclusive
rerun_gc_white_light	remodeling_score	1.379	0.668	-0.710	0.018	0.132	0.925	inconclusive
rerun_gc_white_light	tlr4_innate	0.229	0.149	-0.080	0.220	0.363	0.925	inconclusive
rerun_gc_white_light	fibrosis_tgfb.emt	0.714	0.547	-0.167	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	ecm_organization	0.798	0.723	-0.075	0.010	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	integrin_cell_adhesion	0.869	0.836	-0.033	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	mmp_adam_proteolysis	0.779	0.687	-0.093	0.003	8.75e-04	0.925	tlr4_not_required
rerun_gc_white_light	dct_ncc_wnk_transport	-0.356	-0.106	0.250	0.031	0.506	0.925	inconclusive

S8 Live-return Vector and Component Tables

Table S15: **LAR-versus-ISS-T vector summary.** Complete curated vector-level table from `lar_reversal_vector_summary.tsv`.

Feature set	Age	n	ISS norm	LAR norm	cos	beta	CI low	CI high	perm p	Interpretation
state_space	pooled	4.000	2.245	0.794	-0.564	-0.199	-0.940	0.339	0.177	model_A_attenuation_candidate
state_space	YNG	4.000	2.558	0.590	-0.982	-0.227	-0.969	0.559	0.003	model_A_attenuation_candidate
state_space	OLD	4.000	2.199	1.531	-0.381	-0.265	-0.882	0.421	0.293	model_C_partial_reversal_or_switch
mechanism_scores	pooled	11.000	1.687	0.807	-0.725	-0.347	-0.874	-0.122	0.016	model_B_reversal_candidate
mechanism_scores	YNG	11.000	2.040	1.018	-0.766	-0.382	-0.871	-0.090	0.011	model_B_reversal_candidate
mechanism_scores	OLD	11.000	1.559	0.977	-0.495	-0.310	-0.797	0.264	0.099	model_C_partial_reversal_or_switch
special_clock_slp_rbm3	pooled	12.000	4.695	1.576	-0.273	-0.092	-0.763	0.244	0.297	model_A_attenuation_candidate
special_clock_slp_rbm3	YNG	12.000	5.057	1.688	-0.161	-0.054	-0.753	0.387	0.380	model_A_attenuation_candidate
special_clock_slp_rbm3	OLD	12.000	4.425	1.762	-0.379	-0.151	-0.865	0.277	0.263	model_A_attenuation_candidate
all_gene_expression	pooled	5.00e+03	15.101	10.007	-0.187	-0.124	-0.451	0.169	0.219	mixed_or_inconclusive
all_gene_expression	YNG	5.00e+03	17.180	13.365	-0.145	-0.113	-0.415	0.240	0.281	mixed_or_inconclusive
all_gene_expression	OLD	5.00e+03	16.879	10.290	-0.086	-0.052	-0.390	0.285	0.347	mixed_or_inconclusive
wgcna_module_eigengenes	pooled	21.000	0.783	0.449	-0.489	-0.280	-0.763	-0.080	0.074	model_B_reversal_candidate
wgcna_module_eigengenes	YNG	21.000	0.884	0.527	-0.387	-0.230	-0.723	0.124	0.156	model_A_attenuation_candidate
wgcna_module_eigengenes	OLD	21.000	0.765	0.434	-0.452	-0.256	-0.732	0.063	0.089	model_C_partial_reversal_or_switch
lioness_node_strength_summary	pooled	2.50e+03	12.252	7.691	-0.183	-0.115	NA	NA	NA	exploratory_precomputed_network_summary

Table S16: **LAR model calls by evidence layer.** Model A indicates attenuation, Model B indicates reversal, and Model C indicates switch/partial redirection.

Layer	Model A	Model B	Model C	cos	beta	Interpretation
state_space	True	False	False	-0.564	-0.199	model_A_attenuation_candidate
mechanism_scores	False	True	False	-0.725	-0.347	model_B_reversal_candidate
special_clock_slp_rbm3	True	False	False	-0.273	-0.092	model_A_attenuation_candidate
component:matrix_component	True	False	True	NA	NA	component_arm_divergence
component:dct_transport_component	True	False	False	NA	NA	component_arm_divergence
component:matrix_minus_dct	True	False	True	NA	NA	component_arm_divergence
component:slp_slpr3	True	True	True	NA	NA	component_arm_divergence
component:clock_core	True	False	True	NA	NA	component_arm_divergence
component:rbm3_preservation_expression	False	False	True	NA	NA	component_arm_divergence
mechanism_switch_axis:circadian_dct_state	True	False	True	NA	NA	axis_decomposition
mechanism_switch_axis:immune_tlr4_macrophage	True	True	True	NA	NA	axis_decomposition
mechanism_switch_axis:matrix_high_dct_low	True	True	True	NA	NA	axis_decomposition
mechanism_switch_axis:preservation_cold_stress	False	True	True	NA	NA	axis_decomposition
mechanism_switch_axis:slp_matrix	False	True	True	NA	NA	axis_decomposition
mechanism_switch_axis:transport_rebound	False	False	True	NA	NA	axis_decomposition

Table S17: **Complete LAR component effects.** Full curated component-effect table from `lar_reversal_component_effects.tsv`.

Family	Feature	Arm	Age	Effect	p	q
state_space	matrix_component	ISS-T	pooled	1.250	0.002	0.020
state_space	matrix_component	LAR	pooled	-0.488	0.120	0.278
state_space	matrix_component	ISS-T	YNG	1.788	0.016	0.053
state_space	matrix_component	LAR	YNG	-0.374	0.435	0.543
state_space	matrix_component	ISS-T	OLD	0.712	0.095	0.237
state_space	matrix_component	LAR	OLD	-0.603	0.198	0.396
state_space	dct_transport_component	ISS-T	pooled	-1.004	0.035	0.094
state_space	dct_transport_component	LAR	pooled	-0.377	0.461	0.553
state_space	dct_transport_component	ISS-T	YNG	-1.416	0.022	0.067
state_space	dct_transport_component	LAR	YNG	0.408	0.430	0.543
state_space	dct_transport_component	ISS-T	OLD	-0.593	0.431	0.543
state_space	dct_transport_component	LAR	OLD	-1.161	0.131	0.281
state_space	immune_context_component	ISS-T	pooled	1.158	8.78e-05	0.003
state_space	immune_context_component	LAR	pooled	-0.424	0.229	0.405
state_space	immune_context_component	ISS-T	YNG	0.930	0.002	0.020
state_space	immune_context_component	LAR	YNG	-0.177	0.619	0.663
state_space	immune_context_component	ISS-T	OLD	1.385	0.005	0.025
state_space	immune_context_component	LAR	OLD	-0.672	0.280	0.465
state_space	preservation_stress_component	ISS-T	pooled	1.062	0.007	0.031
state_space	preservation_stress_component	LAR	pooled	-0.265	0.540	0.600
state_space	preservation_stress_component	ISS-T	YNG	0.689	0.295	0.465
state_space	preservation_stress_component	LAR	YNG	-0.104	0.895	0.895
state_space	preservation_stress_component	ISS-T	OLD	1.435	0.013	0.050
state_space	preservation_stress_component	LAR	OLD	-0.426	0.387	0.543
state_space	matrix_minus_dct	ISS-T	pooled	2.254	0.003	0.020
state_space	matrix_minus_dct	LAR	pooled	-0.112	0.865	0.895
state_space	matrix_minus_dct	ISS-T	YNG	3.204	0.005	0.025
state_space	matrix_minus_dct	LAR	YNG	-0.782	0.361	0.542
state_space	matrix_minus_dct	ISS-T	OLD	1.304	0.215	0.403
state_space	matrix_minus_dct	LAR	OLD	0.558	0.533	0.600
mechanism_scores	ecm_organization	ISS-T	pooled	0.859	0.006	0.077
mechanism_scores	ecm_organization	LAR	pooled	-0.545	0.028	0.121
mechanism_scores	ecm_organization	ISS-T	YNG	1.248	0.024	0.115
mechanism_scores	ecm_organization	LAR	YNG	-0.833	0.029	0.121
mechanism_scores	ecm_organization	ISS-T	OLD	0.470	0.114	0.297
mechanism_scores	ecm_organization	LAR	OLD	-0.257	0.452	0.606
mechanism_scores	fibrosis_tgfb_emt	ISS-T	pooled	0.068	0.699	0.796
mechanism_scores	fibrosis_tgfb_emt	LAR	pooled	-0.185	0.117	0.297
mechanism_scores	fibrosis_tgfb_emt	ISS-T	YNG	-0.021	0.934	0.949
mechanism_scores	fibrosis_tgfb_emt	LAR	YNG	-0.146	0.454	0.606
mechanism_scores	fibrosis_tgfb_emt	ISS-T	OLD	0.156	0.538	0.652
mechanism_scores	fibrosis_tgfb_emt	LAR	OLD	-0.223	0.063	0.207
mechanism_scores	tlr4_innate	ISS-T	pooled	0.602	1.23e-05	8.13e-04
mechanism_scores	tlr4_innate	LAR	pooled	-0.223	0.066	0.208
mechanism_scores	tlr4_innate	ISS-T	YNG	0.522	4.00e-04	0.009
mechanism_scores	tlr4_innate	LAR	YNG	-0.073	0.544	0.652
mechanism_scores	tlr4_innate	ISS-T	OLD	0.682	0.008	0.077
mechanism_scores	tlr4_innate	LAR	OLD	-0.372	0.099	0.284
mechanism_scores	integrin_cell_adhesion	ISS-T	pooled	0.445	0.012	0.086
mechanism_scores	integrin_cell_adhesion	LAR	pooled	-0.118	0.456	0.606
mechanism_scores	integrin_cell_adhesion	ISS-T	YNG	0.622	0.034	0.127
mechanism_scores	integrin_cell_adhesion	LAR	YNG	0.063	0.812	0.864
mechanism_scores	integrin_cell_adhesion	ISS-T	OLD	0.268	0.172	0.343

mechanism_scores	integrin_cell_adhesion	LAR	OLD	-0.299	0.147	0.313
mechanism_scores	s1p_s1pr3	ISS-T	pooled	0.727	3.89e-04	0.009
mechanism_scores	s1p_s1pr3	LAR	pooled	-0.271	0.024	0.115
mechanism_scores	s1p_s1pr3	ISS-T	YNG	0.808	0.002	0.028
mechanism_scores	s1p_s1pr3	LAR	YNG	-0.293	0.146	0.313
mechanism_scores	s1p_s1pr3	ISS-T	OLD	0.645	0.054	0.187
mechanism_scores	s1p_s1pr3	LAR	OLD	-0.249	0.135	0.313
mechanism_scores	mmp_adam_proteolysis	ISS-T	pooled	0.535	0.011	0.086
mechanism_scores	mmp_adam_proteolysis	LAR	pooled	-0.139	0.478	0.607
mechanism_scores	mmp_adam_proteolysis	ISS-T	YNG	0.770	0.016	0.097
mechanism_scores	mmp_adam_proteolysis	LAR	YNG	-0.018	0.957	0.957
mechanism_scores	mmp_adam_proteolysis	ISS-T	OLD	0.300	0.318	0.552
mechanism_scores	mmp_adam_proteolysis	LAR	OLD	-0.261	0.315	0.552
mechanism_scores	dct_ncc_wnk_transport	ISS-T	pooled	-0.398	0.035	0.127
mechanism_scores	dct_ncc_wnk_transport	LAR	pooled	-0.149	0.461	0.606
mechanism_scores	dct_ncc_wnk_transport	ISS-T	YNG	-0.561	0.022	0.115
mechanism_scores	dct_ncc_wnk_transport	LAR	YNG	0.162	0.430	0.606
mechanism_scores	dct_ncc_wnk_transport	ISS-T	OLD	-0.235	0.431	0.606
mechanism_scores	dct_ncc_wnk_transport	LAR	OLD	-0.460	0.131	0.313
mechanism_scores	tubular_transport_broad	ISS-T	pooled	-0.267	0.280	0.528
mechanism_scores	tubular_transport_broad	LAR	pooled	-0.167	0.407	0.606
mechanism_scores	tubular_transport_broad	ISS-T	YNG	-0.246	0.456	0.606
mechanism_scores	tubular_transport_broad	LAR	YNG	0.077	0.781	0.845
mechanism_scores	tubular_transport_broad	ISS-T	OLD	-0.289	0.469	0.606
mechanism_scores	tubular_transport_broad	LAR	OLD	-0.411	0.196	0.381
mechanism_scores	oxidative_stress_nrf2	ISS-T	pooled	-0.047	0.871	0.912
mechanism_scores	oxidative_stress_nrf2	LAR	pooled	0.271	0.145	0.313
mechanism_scores	oxidative_stress_nrf2	ISS-T	YNG	-0.356	0.455	0.606
mechanism_scores	oxidative_stress_nrf2	LAR	YNG	0.425	0.166	0.343
mechanism_scores	oxidative_stress_nrf2	ISS-T	OLD	0.263	0.388	0.606
mechanism_scores	oxidative_stress_nrf2	LAR	OLD	0.118	0.641	0.756
mechanism_scores	macrophage_inflammation	ISS-T	pooled	0.380	0.072	0.216
mechanism_scores	macrophage_inflammation	LAR	pooled	-0.135	0.658	0.761
mechanism_scores	macrophage_inflammation	ISS-T	YNG	0.220	0.369	0.606
mechanism_scores	macrophage_inflammation	LAR	YNG	-0.100	0.772	0.845
mechanism_scores	macrophage_inflammation	ISS-T	OLD	0.539	0.106	0.292
mechanism_scores	macrophage_inflammation	LAR	OLD	-0.170	0.741	0.829
mechanism_scores	preservation_stress_response	ISS-T	pooled	0.594	0.007	0.077
mechanism_scores	preservation_stress_response	LAR	pooled	-0.148	0.540	0.652
mechanism_scores	preservation_stress_response	ISS-T	YNG	0.385	0.295	0.540
mechanism_scores	preservation_stress_response	LAR	YNG	-0.058	0.895	0.923
mechanism_scores	preservation_stress_response	ISS-T	OLD	0.802	0.013	0.088
mechanism_scores	preservation_stress_response	LAR	OLD	-0.238	0.387	0.606
special_clock_s1p_rbm3	clock_core	ISS-T	pooled	-1.052	0.001	0.014
special_clock_s1p_rbm3	clock_core	LAR	pooled	0.118	0.420	0.573
special_clock_s1p_rbm3	clock_core	ISS-T	YNG	-1.038	0.014	0.052
special_clock_s1p_rbm3	clock_core	LAR	YNG	0.018	0.916	0.936
special_clock_s1p_rbm3	clock_core	ISS-T	OLD	-1.065	0.054	0.145
special_clock_s1p_rbm3	clock_core	LAR	OLD	0.218	0.388	0.558
special_clock_s1p_rbm3	per_cry	ISS-T	pooled	-2.020	5.96e-05	0.002
special_clock_s1p_rbm3	per_cry	LAR	pooled	0.041	0.861	0.923
special_clock_s1p_rbm3	per_cry	ISS-T	YNG	-2.001	0.004	0.029
special_clock_s1p_rbm3	per_cry	LAR	YNG	-0.142	0.658	0.789
special_clock_s1p_rbm3	per_cry	ISS-T	OLD	-2.039	0.014	0.052
special_clock_s1p_rbm3	per_cry	LAR	OLD	0.224	0.554	0.687
special_clock_s1p_rbm3	bmal_clock	ISS-T	pooled	1.680	2.84e-04	0.005
special_clock_s1p_rbm3	bmal_clock	LAR	pooled	0.079	0.747	0.854
special_clock_s1p_rbm3	bmal_clock	ISS-T	YNG	1.406	0.028	0.087

special_clock_s1p_rbm3	bmal_clock	LAR	YNG	0.504	0.138	0.283
special_clock_s1p_rbm3	bmal_clock	ISS-T	OLD	1.954	0.012	0.052
special_clock_s1p_rbm3	bmal_clock	LAR	OLD	-0.345	0.362	0.531
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	pooled	-0.450	0.066	0.164
special_clock_s1p_rbm3	dct_wnk_expression	LAR	pooled	-0.016	0.936	0.936
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	YNG	-0.750	0.014	0.052
special_clock_s1p_rbm3	dct_wnk_expression	LAR	YNG	0.228	0.325	0.517
special_clock_s1p_rbm3	dct_wnk_expression	ISS-T	OLD	-0.150	0.690	0.815
special_clock_s1p_rbm3	dct_wnk_expression	LAR	OLD	-0.260	0.356	0.531
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	pooled	-0.334	0.190	0.333
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	pooled	-0.418	0.148	0.288
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	YNG	-0.251	0.478	0.604
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	YNG	-0.068	0.806	0.879
special_clock_s1p_rbm3	aldosterone_enac_expression	ISS-T	OLD	-0.417	0.294	0.482
special_clock_s1p_rbm3	aldosterone_enac_expression	LAR	OLD	-0.769	0.146	0.288
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	pooled	0.452	0.004	0.029
special_clock_s1p_rbm3	s1p_axis_expression	LAR	pooled	-0.269	0.014	0.052
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	YNG	0.521	0.011	0.052
special_clock_s1p_rbm3	s1p_axis_expression	LAR	YNG	-0.222	0.174	0.312
special_clock_s1p_rbm3	s1p_axis_expression	ISS-T	OLD	0.383	0.100	0.231
special_clock_s1p_rbm3	s1p_axis_expression	LAR	OLD	-0.316	0.054	0.145
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	pooled	0.501	0.004	0.029
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	pooled	-0.252	0.245	0.421
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	YNG	0.409	0.153	0.288
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	YNG	-0.256	0.464	0.604
special_clock_s1p_rbm3	rbm3_preservation_expression	ISS-T	OLD	0.593	0.022	0.076
special_clock_s1p_rbm3	rbm3_preservation_expression	LAR	OLD	-0.249	0.430	0.573
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	pooled	-1.876	7.85e-05	0.002
special_clock_s1p_rbm3	Rbm3_expression	LAR	pooled	0.352	0.350	0.531
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	YNG	-2.278	0.002	0.014
special_clock_s1p_rbm3	Rbm3_expression	LAR	YNG	0.173	0.799	0.879
special_clock_s1p_rbm3	Rbm3_expression	ISS-T	OLD	-1.475	0.038	0.113
special_clock_s1p_rbm3	Rbm3_expression	LAR	OLD	0.530	0.156	0.288
special_clock_s1p_rbm3	Per2_expression	ISS-T	pooled	-2.527	8.77e-05	0.002
special_clock_s1p_rbm3	Per2_expression	LAR	pooled	0.036	0.890	0.928
special_clock_s1p_rbm3	Per2_expression	ISS-T	YNG	-2.705	0.008	0.044
special_clock_s1p_rbm3	Per2_expression	LAR	YNG	-0.089	0.786	0.879
special_clock_s1p_rbm3	Per2_expression	ISS-T	OLD	-2.349	0.012	0.052
special_clock_s1p_rbm3	Per2_expression	LAR	OLD	0.161	0.708	0.822
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	pooled	1.276	0.001	0.014
special_clock_s1p_rbm3	S1pr3_expression	LAR	pooled	-1.269	0.002	0.014
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	YNG	1.572	0.008	0.044
special_clock_s1p_rbm3	S1pr3_expression	LAR	YNG	-1.401	0.028	0.087
special_clock_s1p_rbm3	S1pr3_expression	ISS-T	OLD	0.980	0.109	0.245
special_clock_s1p_rbm3	S1pr3_expression	LAR	OLD	-1.136	0.047	0.134
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	pooled	-1.114	0.060	0.154
special_clock_s1p_rbm3	Slc12a3_expression	LAR	pooled	-0.074	0.872	0.923
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	YNG	-1.373	0.079	0.189
special_clock_s1p_rbm3	Slc12a3_expression	LAR	YNG	0.069	0.926	0.936
special_clock_s1p_rbm3	Slc12a3_expression	ISS-T	OLD	-0.855	0.330	0.517
special_clock_s1p_rbm3	Slc12a3_expression	LAR	OLD	-0.218	0.644	0.786
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	pooled	-0.695	0.130	0.283
special_clock_s1p_rbm3	Wnk4_expression	LAR	pooled	-0.641	0.272	0.456
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	YNG	-0.761	0.137	0.283
special_clock_s1p_rbm3	Wnk4_expression	LAR	YNG	-0.631	0.475	0.604
special_clock_s1p_rbm3	Wnk4_expression	ISS-T	OLD	-0.629	0.424	0.573
special_clock_s1p_rbm3	Wnk4_expression	LAR	OLD	-0.650	0.419	0.573

S9 Figure Gallery

S9.1 Main Manuscript Figures

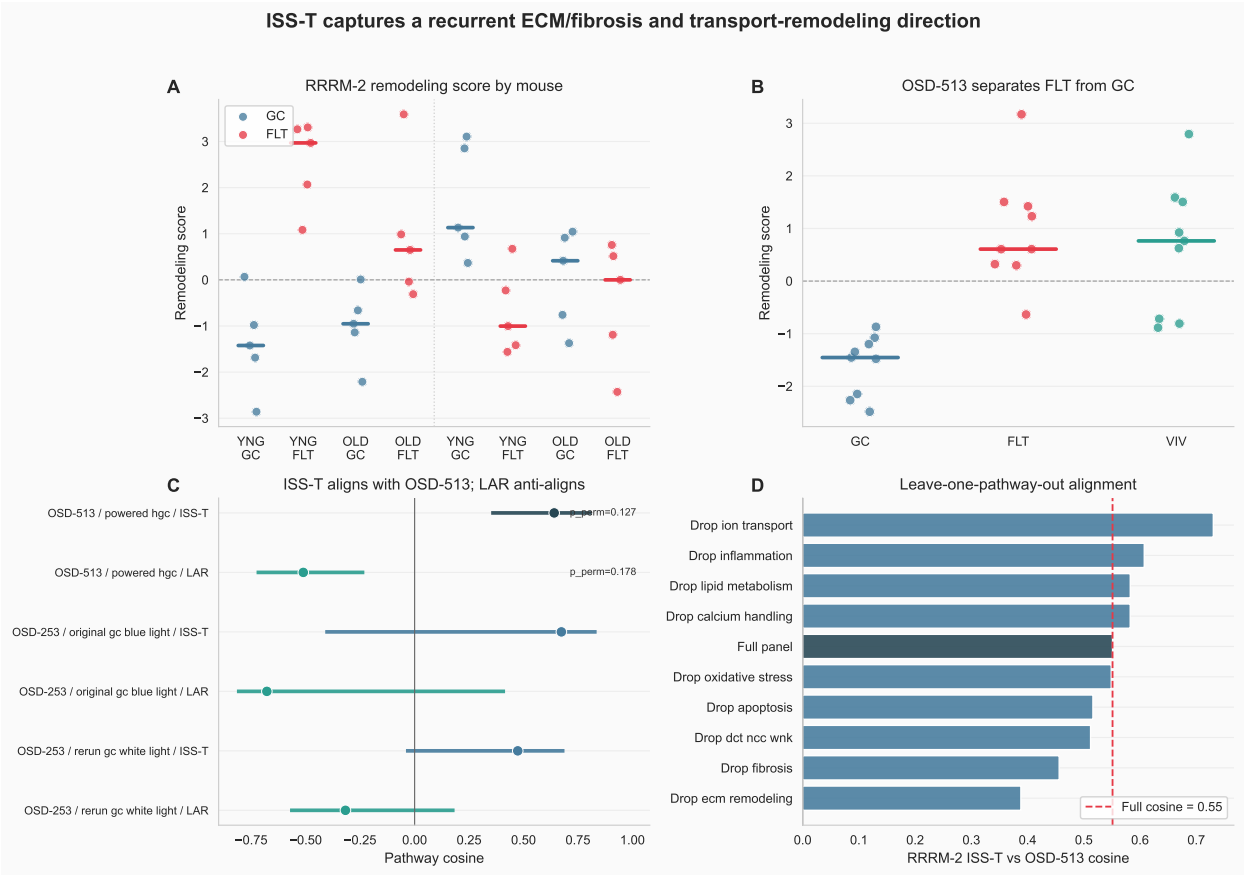


Figure S1: **Main result overview.** RRRM-2 and OSD-513 remodeling scores, cross-OSDR alignment, and leave-one-pathway-out sensitivity.

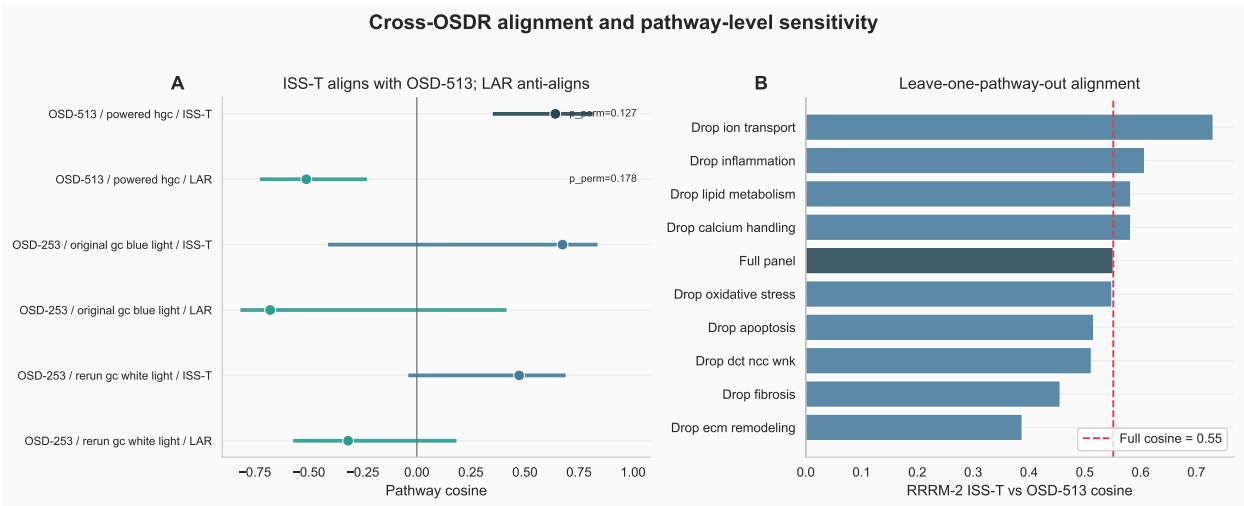


Figure S2: **Cross-OSDR alignment and leave-one-pathway-out sensitivity.** ISS-T aligns with OSD-513; LAR is directionally anti-aligned.

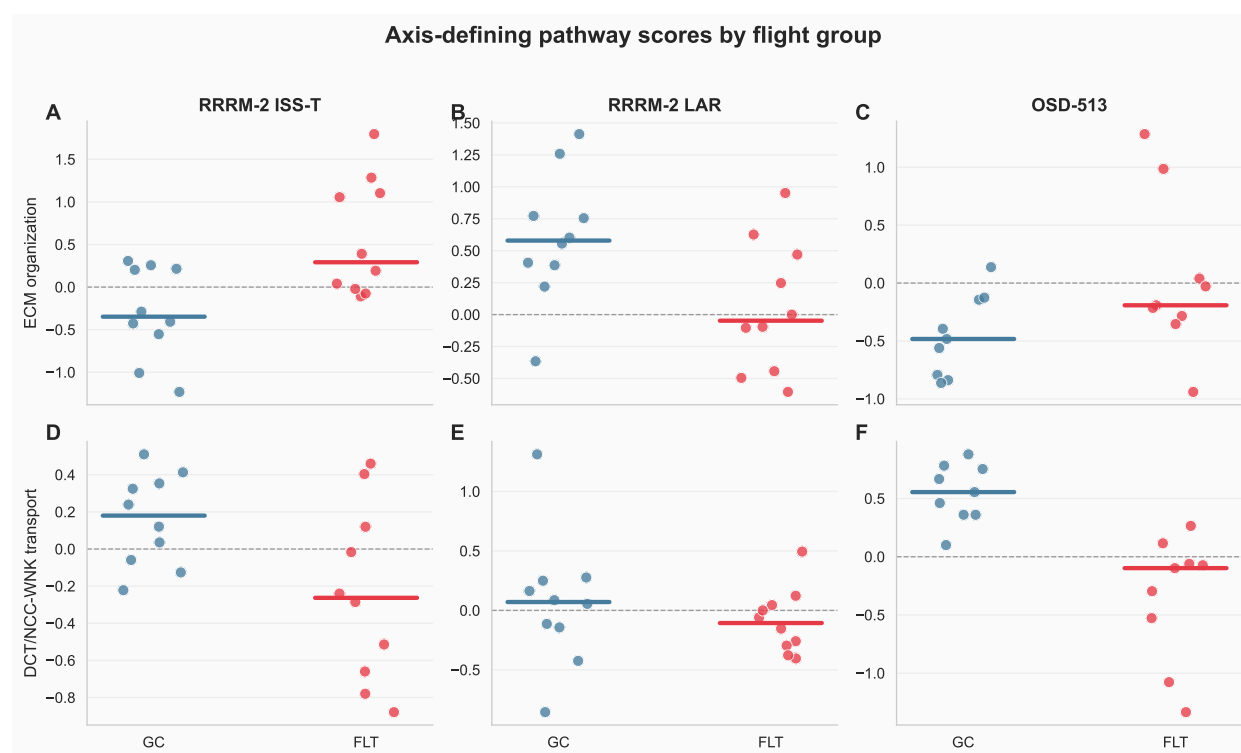


Figure S3: **Axis-defining pathway scores by flight group.** ECM/stromal features rise and DCT/NCC-WNK transport falls in the aligned terminal-flight contexts.

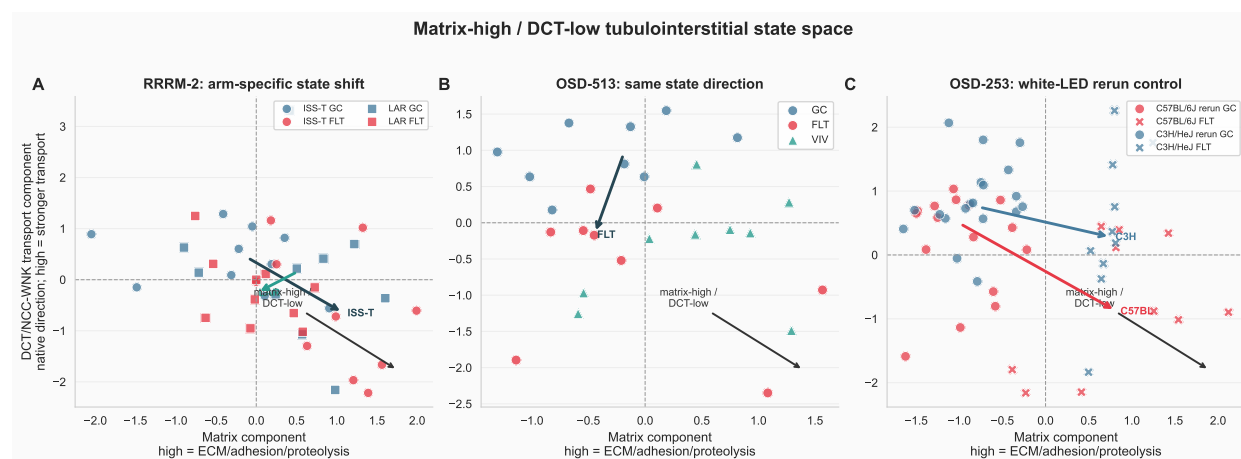


Figure S4: **DCT/NCC-low state space.** Matrix component on the x-axis, DCT transport in native direction on the y-axis.

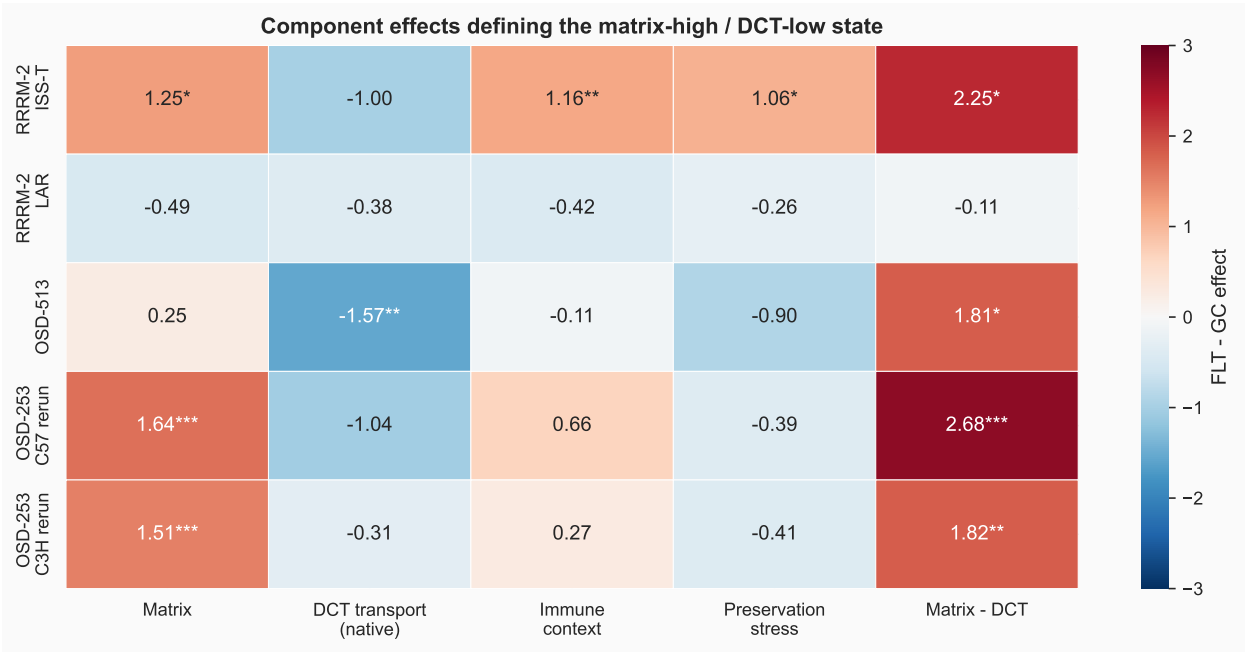


Figure S5: **State component effects.** FLT-minus-GC effects in standardized component units; DCT transport is shown in native direction.



Figure S6: **RRRM-2 deconvolution overlay.** The state score and components do not reduce to estimated DCT or fibroblast fraction in the available deconvolution model.

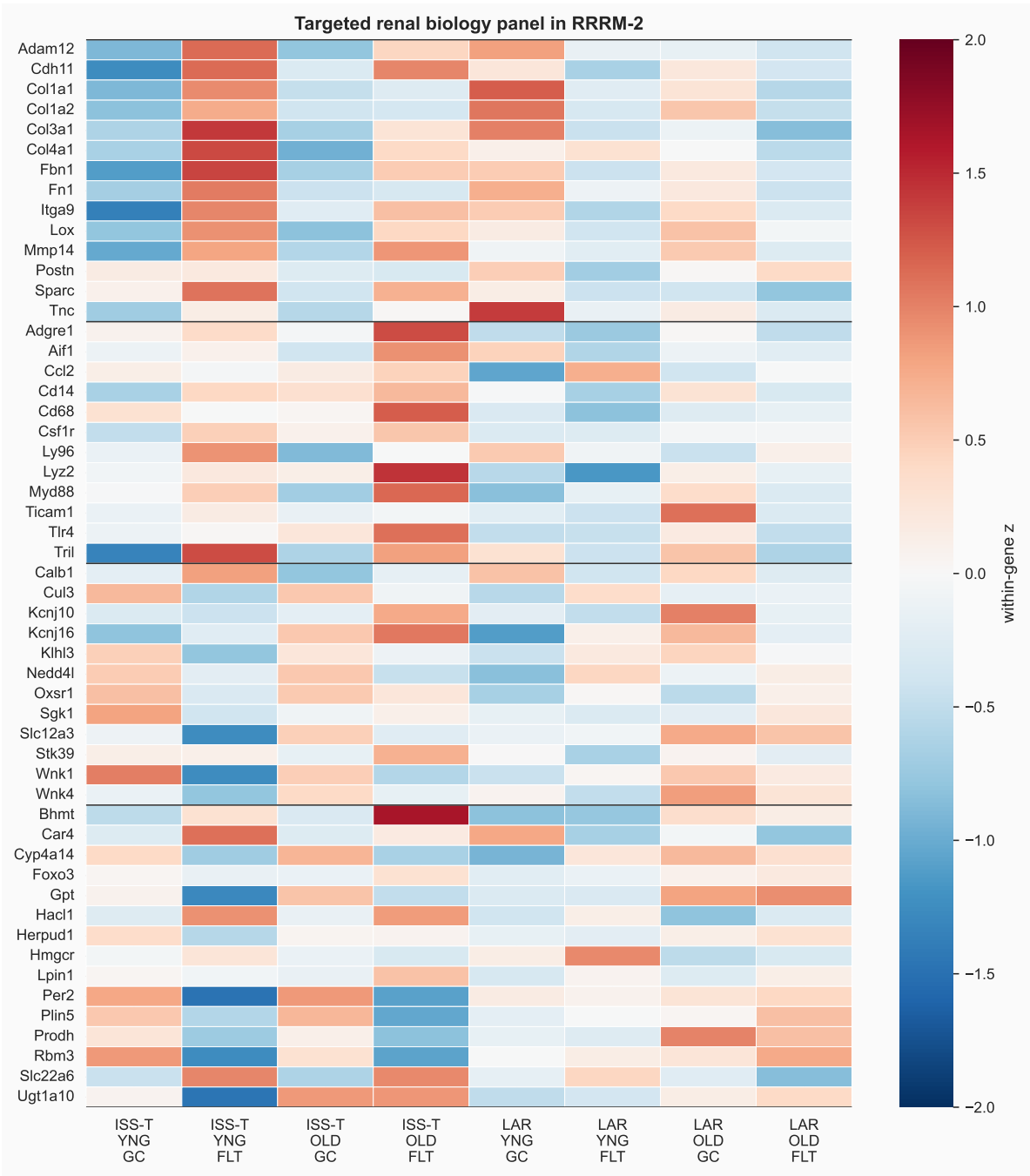


Figure S7: **Targeted renal biology panel.** Residualized RRRM-2 gene expression for ECM/adhesion/proteolysis, innate/macrophage, DCT/NCC-WNK, and tubular/metabolic genes.

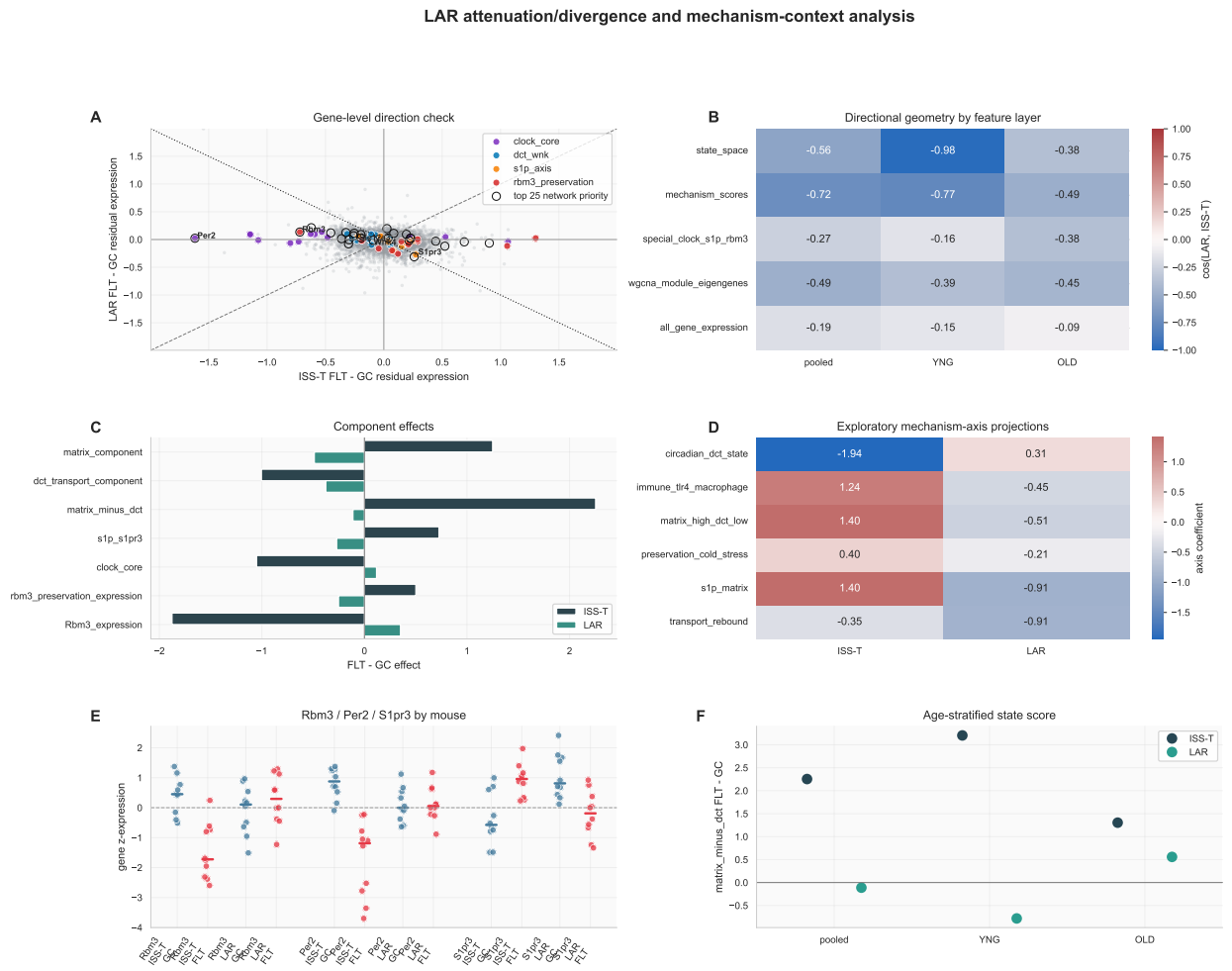


Figure S8: **Live-return arm attenuation/divergence dashboard.** LAR supports attenuation/redirection rather than confirmed reversal.

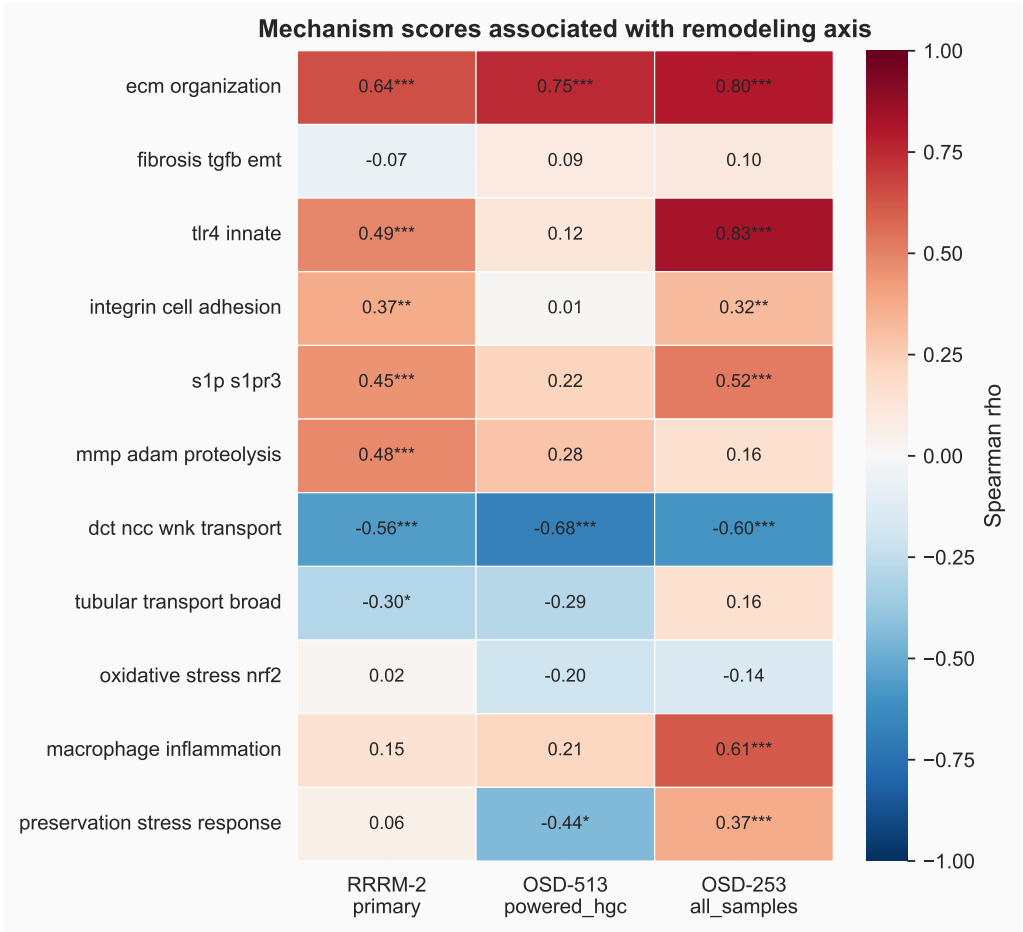


Figure S9: **Mechanism-score associations with the recurrent state** across RRRM-2, OSD-513, and OSD-253.

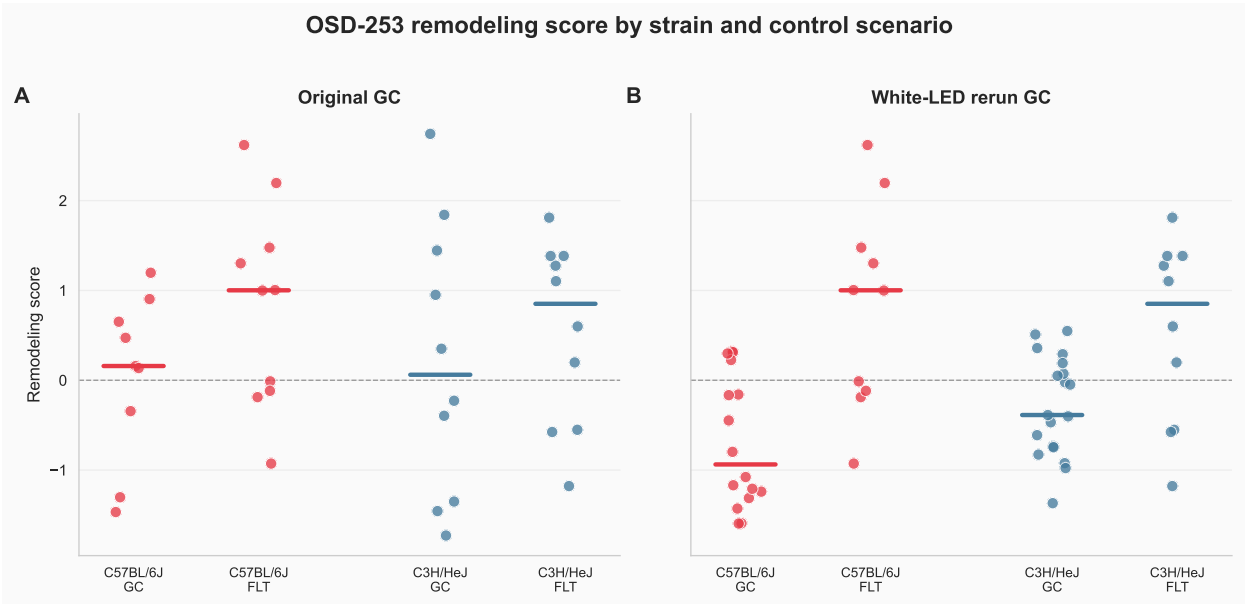


Figure S10: **OSD-253 strain/control-context sensitivity.** C57BL/6J and C3H/HeJ state scores under original and rerun ground-control scenarios.

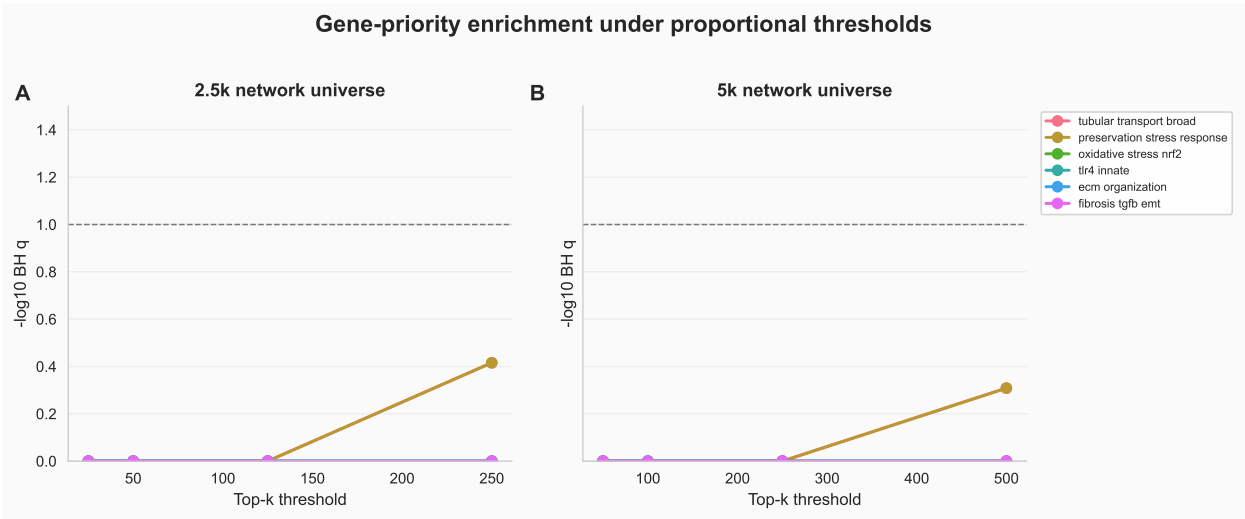


Figure S11: **LIONESS/node2vec gene-priority sensitivity.** Network-prioritized genes provide candidate neighborhoods but do not validate at protein/phosphoprotein level.

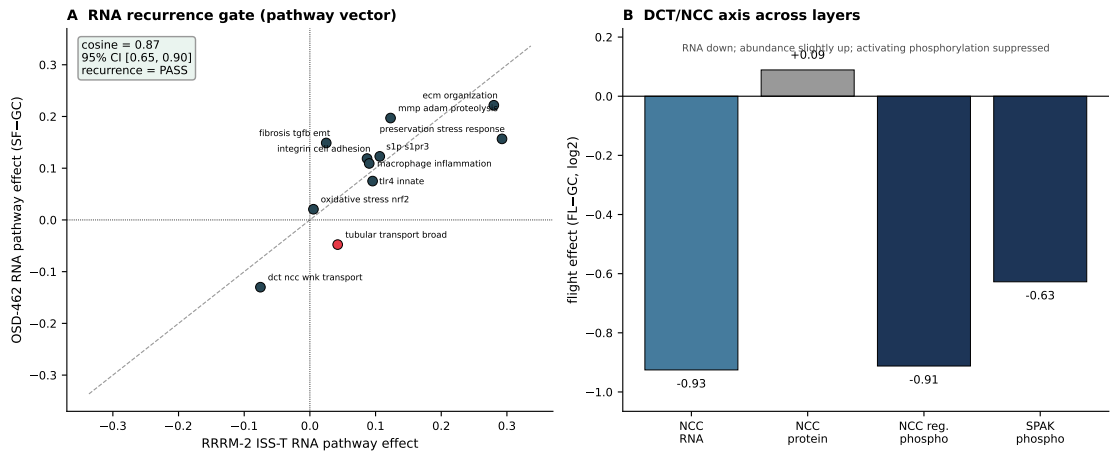


Figure S12: **OSD-462 RNA recurrence gate.** OSD-462 RNA reproduces the RRRM-2 ISS-T DCT/NCC-low context direction.

OSD-462 / RR-10 multi-omics anchor — protein & phosphoprotein test of the RRRM-2 matrix-high / DCT-low remodeling signal

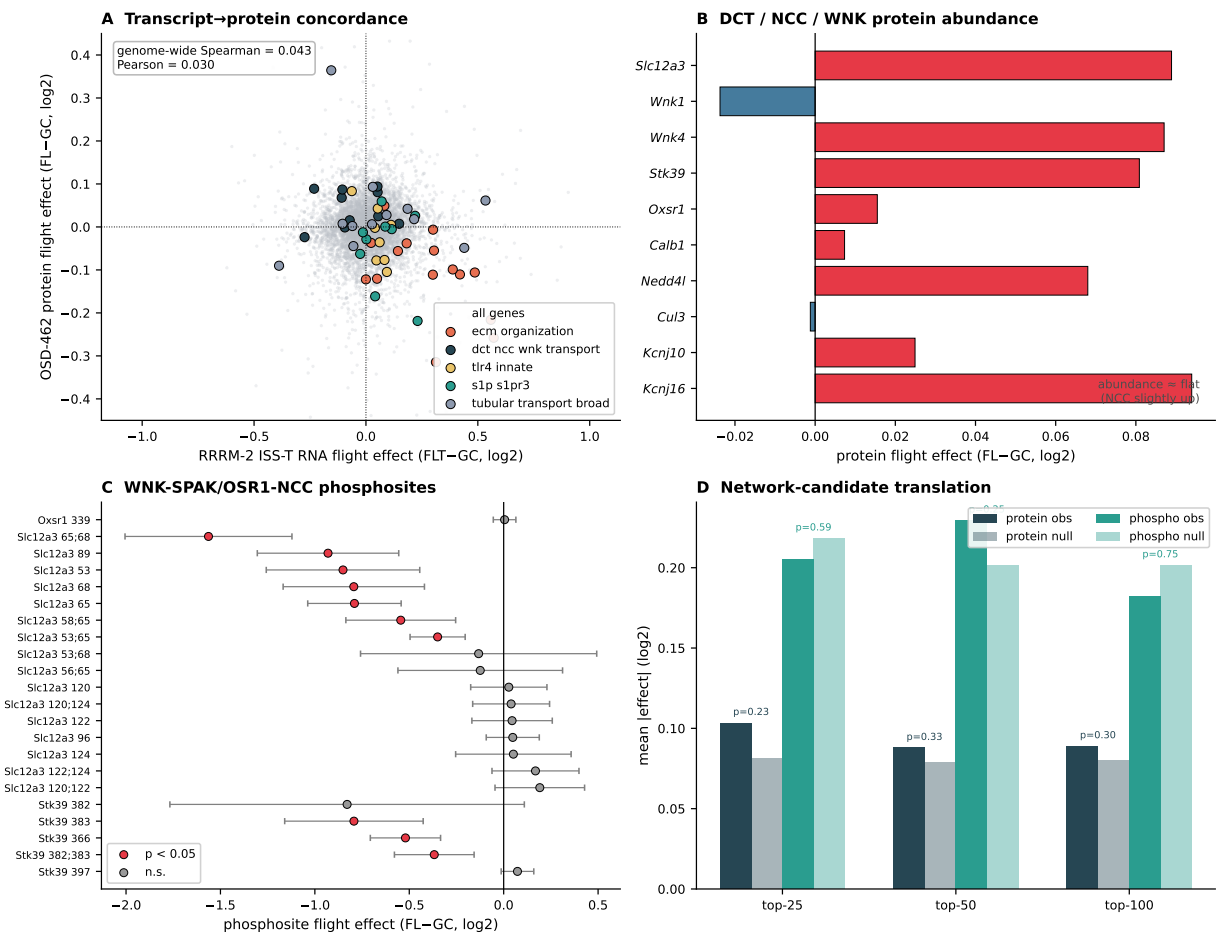
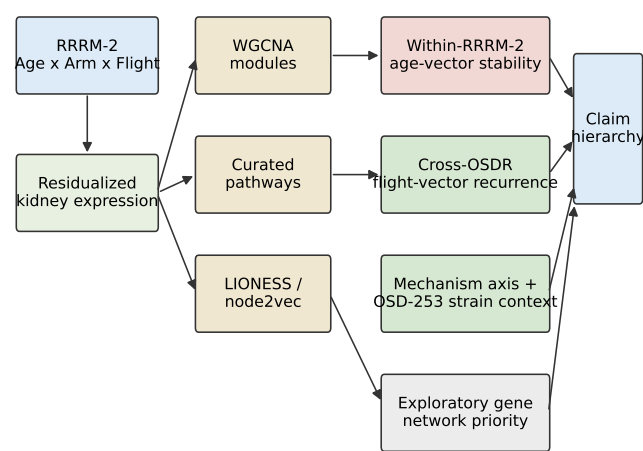


Figure S13: **OSD-462 multi-omic anchor dashboard.** Transcript-level recurrence, protein-abundance null, and phospho-axis anchoring of the established NCC/SPAK phenotype.

S9.2 Secondary and Legacy Curated Figures



Primary claim: cross-cohort ISS-T-aligned flight remodeling. Boundary: no broad RRRM-2 age-rewiring claim; gene rankings remain exploratory.

Figure S14: **Workflow and claim hierarchy.** Analysis layers and claim boundaries used for the state-map manuscript.

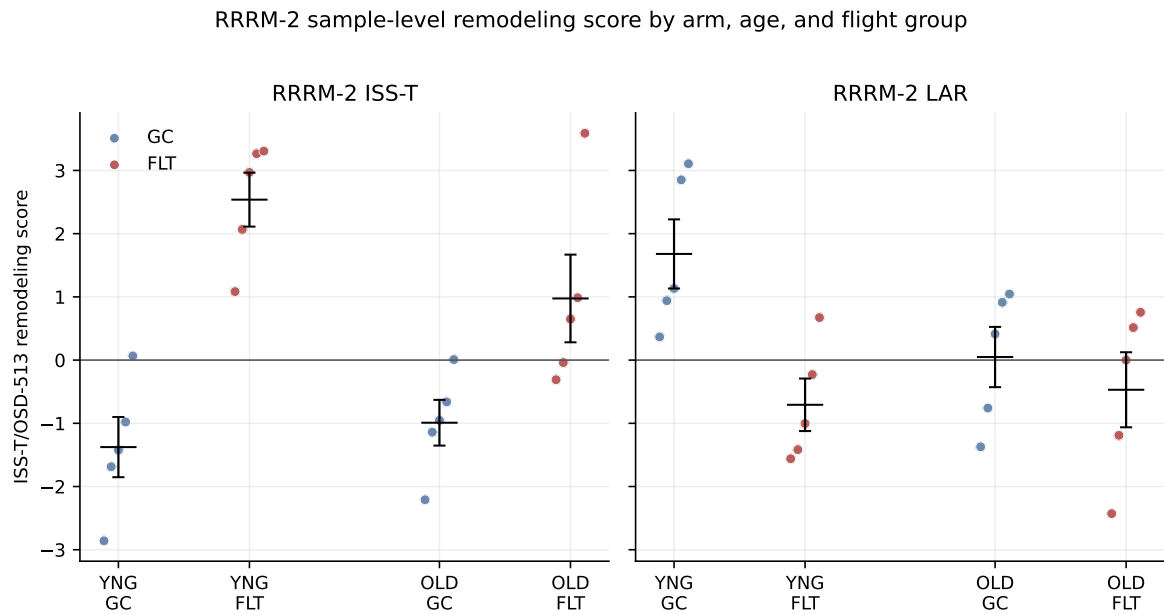


Figure S15: **RRRM-2 remodeling points.** Earlier standalone view of RRRM-2 remodeling scores by group.

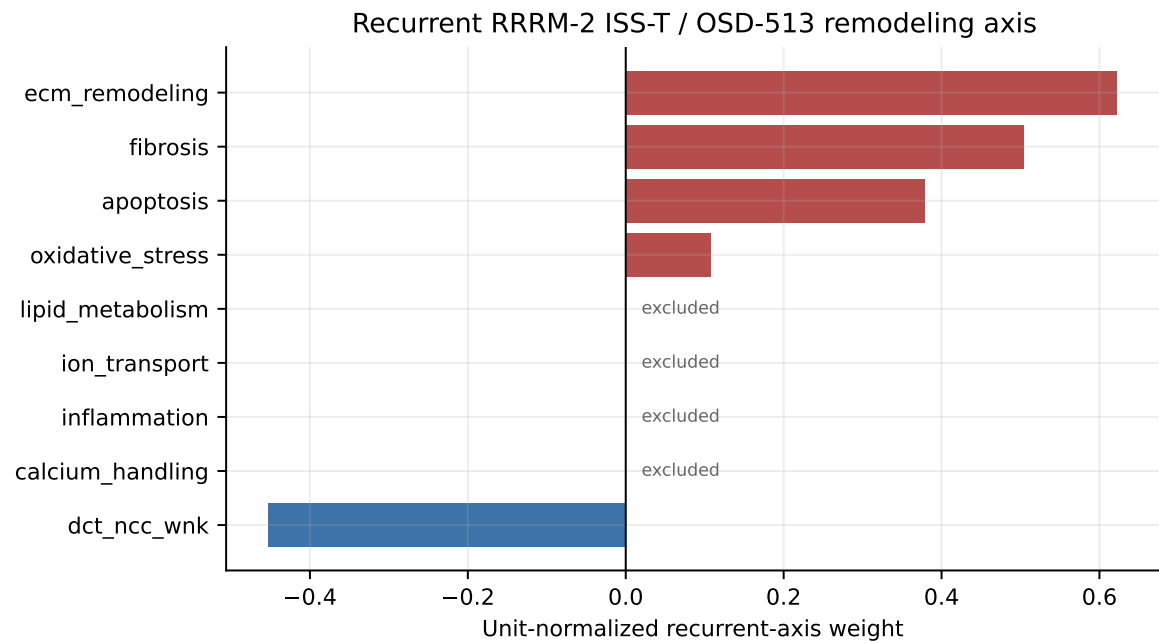


Figure S16: **Recurrent-axis weights.** Pathway weights defining the terminal-flight DCT/NCC-low context direction.

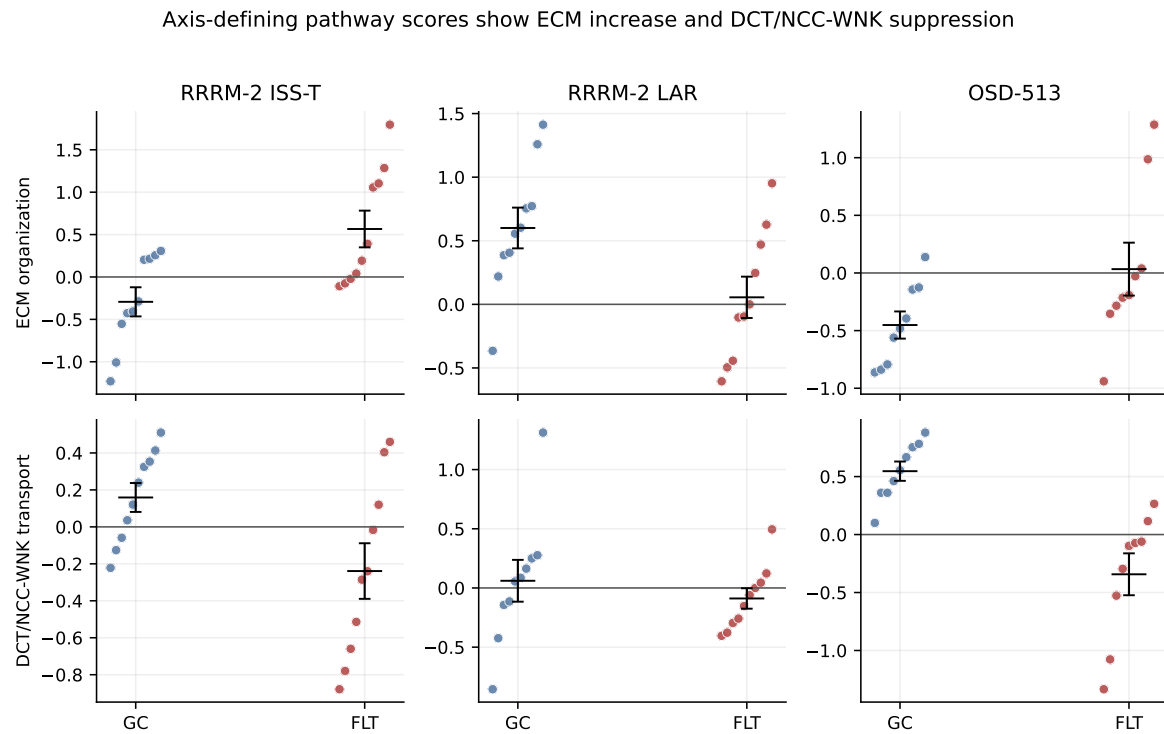


Figure S17: **Pathway scores by flight, alternate numbering.** Earlier figure export retained for completeness.

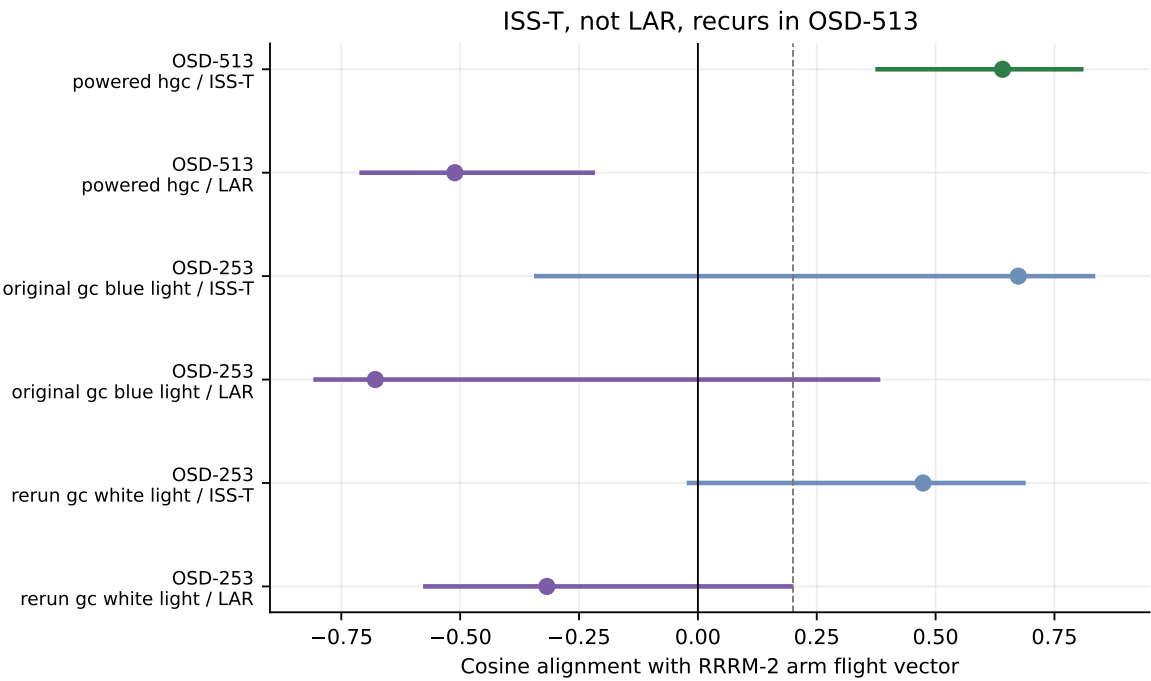


Figure S18: **Cross-OSDR alignment, standalone view.** Earlier figure export retained for completeness.

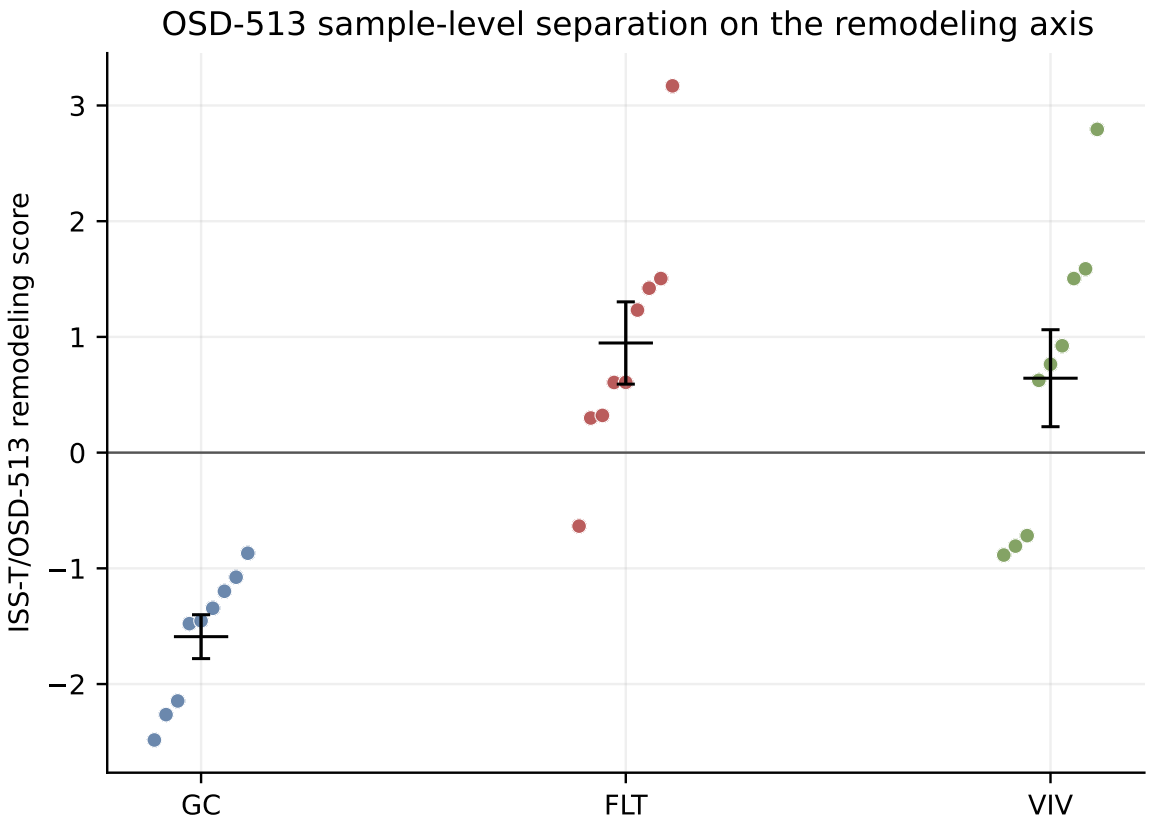


Figure S19: **OSD-513 sample separation.** Independent kidney cohort state-score separation.

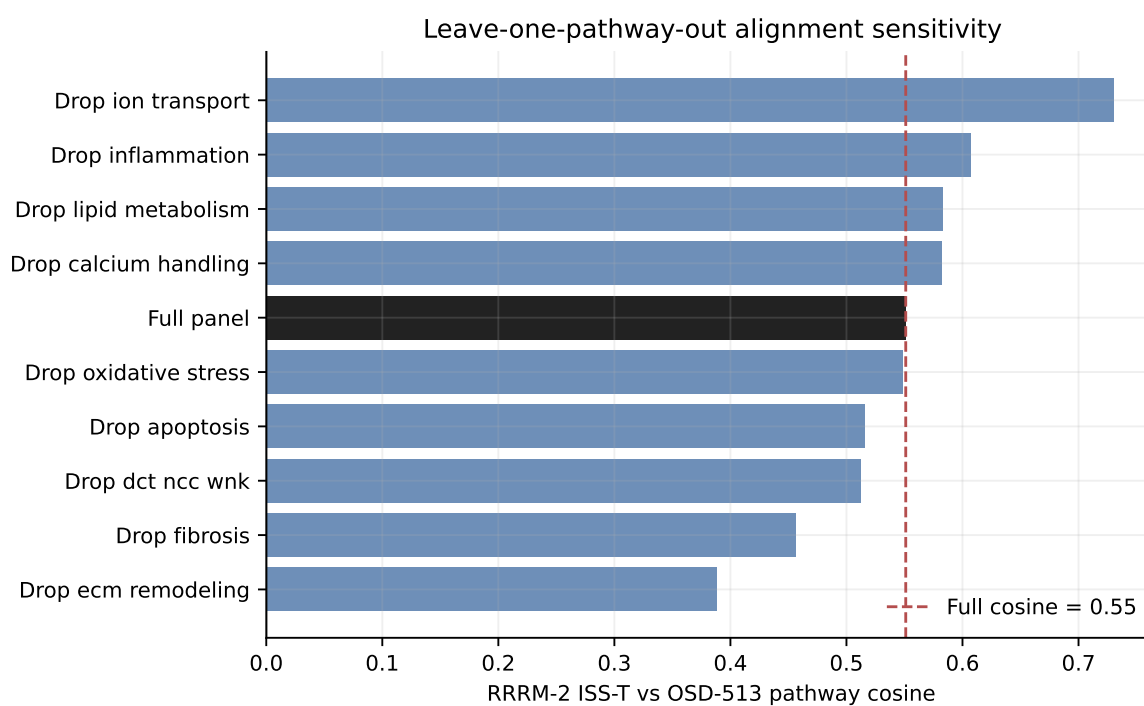


Figure S20: **Leave-one-pathway-out sensitivity.** Earlier standalone sensitivity view.

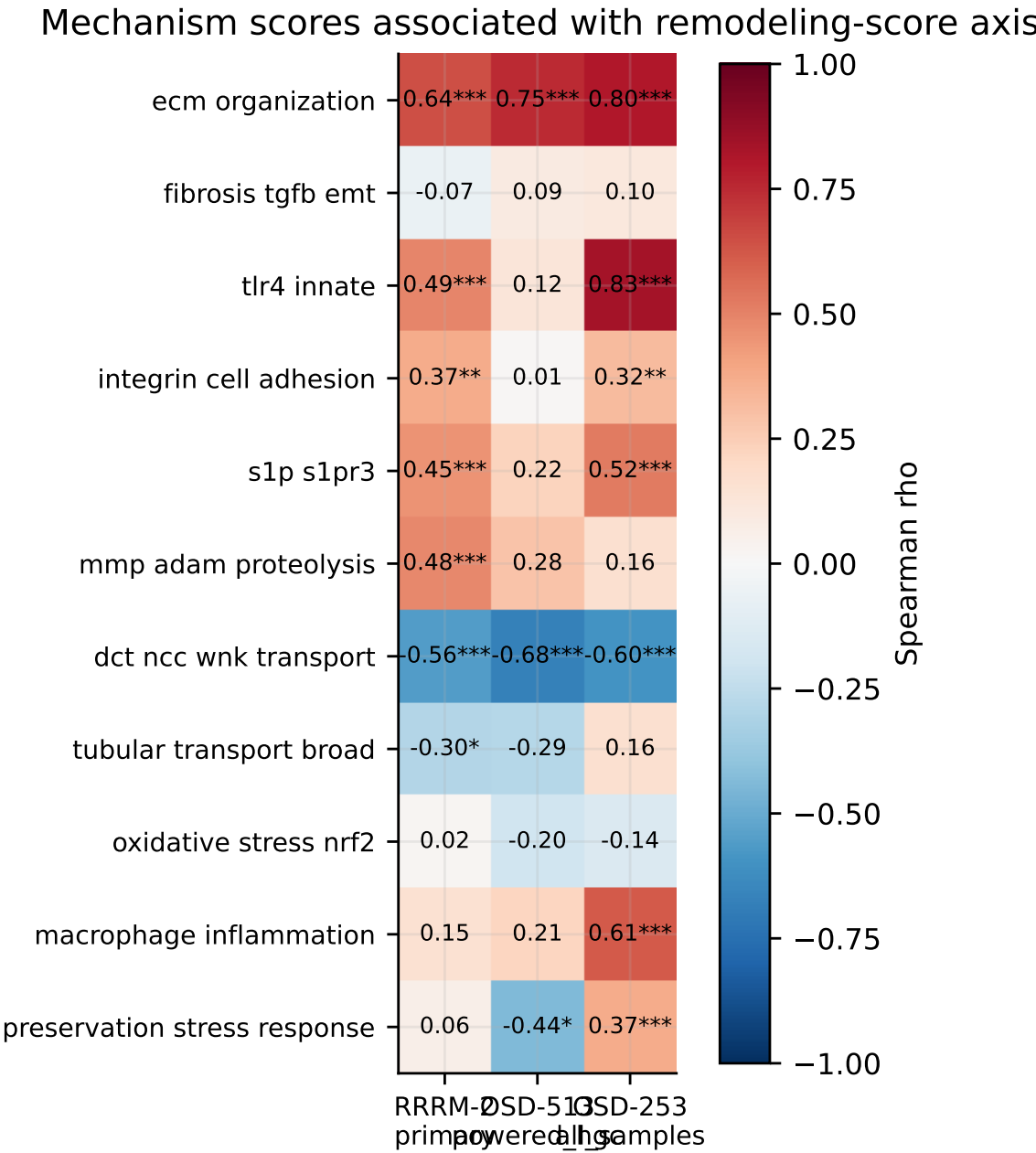


Figure S21: Mechanism-score heatmap, earlier export. Retained for completeness.

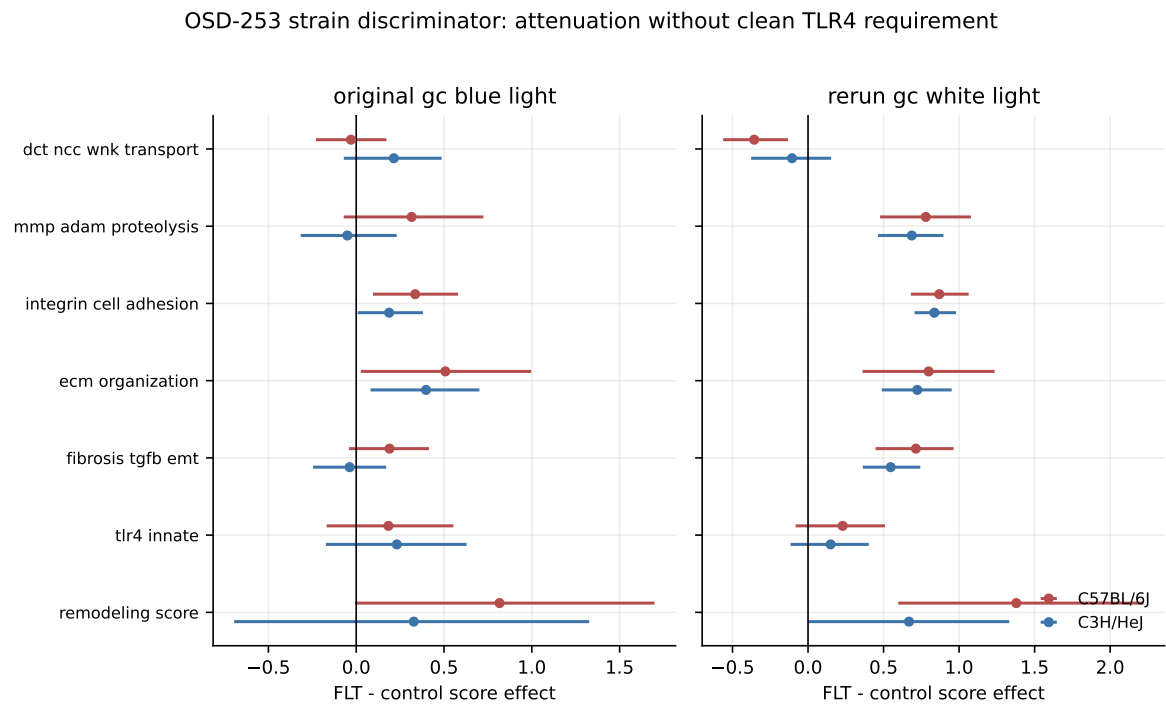


Figure S22: **OSD-253 strain discriminator.** Earlier TLR4/strain-context visualization.

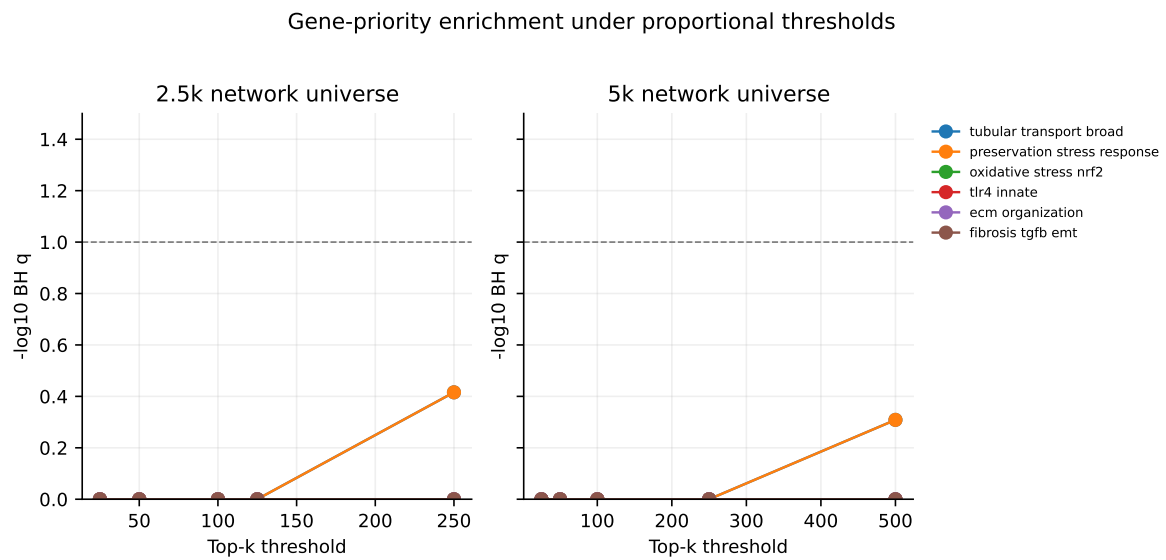


Figure S23: **Gene-priority sensitivity, earlier export.** Retained for completeness.

S10 Complete Artifact Index

Table S18: **Complete generated result artifacts indexed by source path.** The compendium embeds the curated manuscript tables and figures above; these files contain the complete machine-readable outputs used to generate them.

Category	Complete source artifact(s)
Run provenance	data/results/run_20260519_000547_2500g/run_metadata.json; data/results/run_20260519_cosine_perm_null/run_metadata.json; data/results/run_20260519_5000g_sensitivity_3seed/run_metadata.json; data/results/run_20260521_osd462_anchor/osd462_anchor/manifest.json
Age-vector stability	data/results/run_20260519_000547_2500g/contrast_vectors/agc_stability_report.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/agc_stability_decision.json
Cross-OSDR recurrence	data/results/run_20260519_000547_2500g/contrast_vectors/cross_osdr_recurrence/cross_osdr_alignment_summary.tsv; data/results/run_20260519_cosine_perm_null/contrast_vectors/cross_osdr_recurrence/*_pathway_permutation.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/cross_osdr_recurrence/*_pathway_bootstrap.tsv
Recurrent state axis	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/recurrent_axis_weights.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/sample_remodeling_scores.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/mechanism_score_matrix.tsv
Mechanism context	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/mechanism_convergence_summary.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/mechanism_score_associations.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/mechanism_gene_set_coverage.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/axis_gene_set_coverage.tsv
State space	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/state_space_scores.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/state_space_group_effects.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/state_space_component_correlations.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/targeted_renal_gene_panel.tsv

Category	Complete source artifact(s)
LAR attenuation/divergence	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_reversal_vector_summary.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_reversal_component_effects.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_reversal_model_calls.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_s1p_axis_analysis.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_circadian_dct_analysis.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal/lar_rbm3_preservation_analysis.tsv
OSD-253 strain/TLR4 context	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/osd253_tlr4_discriminator.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/osd253_tlr4_discriminator_bootstrap.tsv
Gene/network priority	data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_axis_priority.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_axis_priority_top25.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_axis_priority_top50.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_axis_priority_top100.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_priority_enrichment_topk.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/gene_priority_enrichment_preranked.tsv
External aging	data/results/run_20260519_000547_2500g/contrast_vectors/external_aging_axis/ISS-T/point_estimates.json; data/results/run_20260519_000547_2500g/contrast_vectors/external_aging_axis/LAR/point_estimates.json; data/results/run_20260519_000547_2500g/contrast_vectors/external_aging_axis/ISS-T/bootstrap.tsv; data/results/run_20260519_000547_2500g/contrast_vectors/external_aging_axis/LAR/bootstrap.tsv
OSD-462 RNA anchor	data/results/run_20260521_osd462_anchor/osd462_anchor/osd462_rna_recurrence.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/osd462_rna_recurrence_bootstrap.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/osd462_rna_recurrence_loo.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/osd462_rna_pathway_effects.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/osd462_pathway_coverage.tsv
OSD-462 protein anchor	data/results/run_20260521_osd462_anchor/osd462_anchor/protein_concordance_summary.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/protein_concordance_target_genes.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/protein_concordance_background.tsv; data/results/run_20260521_osd462_anchor/osd462_anchor/protein_effects_gene_anyplex.tsv

Category	Complete source artifact(s)
OSD-462 phosphoprotein anchor	<code>data/results/run_20260521_osd462_anchor/osd462_anchor/phospho_axis_summary.tsv</code> ; <code>data/results/run_20260521_osd462_anchor/osd462_anchor/phospho_axis_coverage.tsv</code> ; <code>data/results/run_20260521_osd462_anchor/osd462_anchor/phospho_axis_verdict.tsv</code> ; <code>data/results/run_20260521_osd462_anchor/osd462_anchor/phospho_all_sites.tsv</code>
OSD-462 network translation	<code>data/results/run_20260521_osd462_anchor/osd462_anchor/network_translation.tsv</code> ; <code>data/results/run_20260521_osd462_anchor/osd462_anchor/network_translation_candidates.tsv</code>
Figures	<code>latex_paper/figures_v4/*.pdf</code> ; <code>latex_paper/figures_osd462/*.pdf</code>

S11 Verification notes

The following independent checks were performed on the OSD-462 anchor. (i) The eight `tests/test_osd462_anchor.py` unit tests pass. (ii) NCC phosphosite flight effects were re-derived directly from the raw `siteQuant_360` workbook ($\log_2$ of TMT signal-to-noise, 10 FL vs 10 GC channels, Welch t -test, no pipeline code): Thr53 -0.99 , Thr65 -0.93 ($p = 2.3 \times 10^{-6}$), Thr68 -0.94 , Thr89 -1.07 ; non-regulatory Ser96/Ser120/Ser122 flat — confirming the pipeline output in direction, magnitude, and significance. (iii) Total NCC protein flight effect ($+0.089$, 70 peptides) and the matrix-protein effects were confirmed from `protein_effects_gene_anyplex.tsv`. (iv) The RNA recurrence cosine and leave-one-pathway-out range were reproduced from `osd462_rna_recurrence.tsv`. The site-specific pattern (regulatory cluster down, non-regulatory sites flat, total protein flat) is the principal internal control establishing that the phospho-axis result is genuine activating-site regulation rather than a normalization or abundance artifact.

Curated results compendium for Manuscript v9. The spaceflight NCC dephosphorylation phenotype anchored here was first established by the Cosmic Kidney Disease study (Siew et al., Nature Communications, 2024); v9 maps the transcriptomic context around that established phenotype.